Medical Physics

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Chapter 11 : Quantum physics

Chapter 11： Quantum physics

You will learn:

1. Plank’s quantum hypothesis
2. Photoelectric effect
3. The Compton effect
4. The Bohr hydrogen atom
5. De Broglie’s hypothesis
6. Radioactivity

Chapter 11： Quantum physics

$E = nh\nu$

$h = 6.626 \times 10^{-34}\,\mathrm{J \cdot s}$

Photoelectric effect

$$h\nu = \frac{1}{2}m\mathrm{v}^2 + E_0$$

$$p = h/\lambda$$

Compton shift

$$\Delta\lambda = \frac{h}{m_0 c}(1 - \cos\theta) = \lambda_C(1 - \cos\theta)$$

Wavelength of matter wave

$$\lambda = \frac{h}{p} = \frac{h}{m\mathrm{v}} = \frac{h}{\sqrt{2mk}}$$

The Uncertainty Principle

$$\Delta \mathbf{y} \bullet \Delta \mathbf{p}_y \geq \mathrm{h}$$

Radioactivity

$$N = N_0 \mathbf{e}^{-\lambda t}$$

Chapter 11： Quantum physics

Lyman series $\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$ n=2, 3, …… Ultraviolet

Balmer series $\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$ n=3, 4, …… Visible

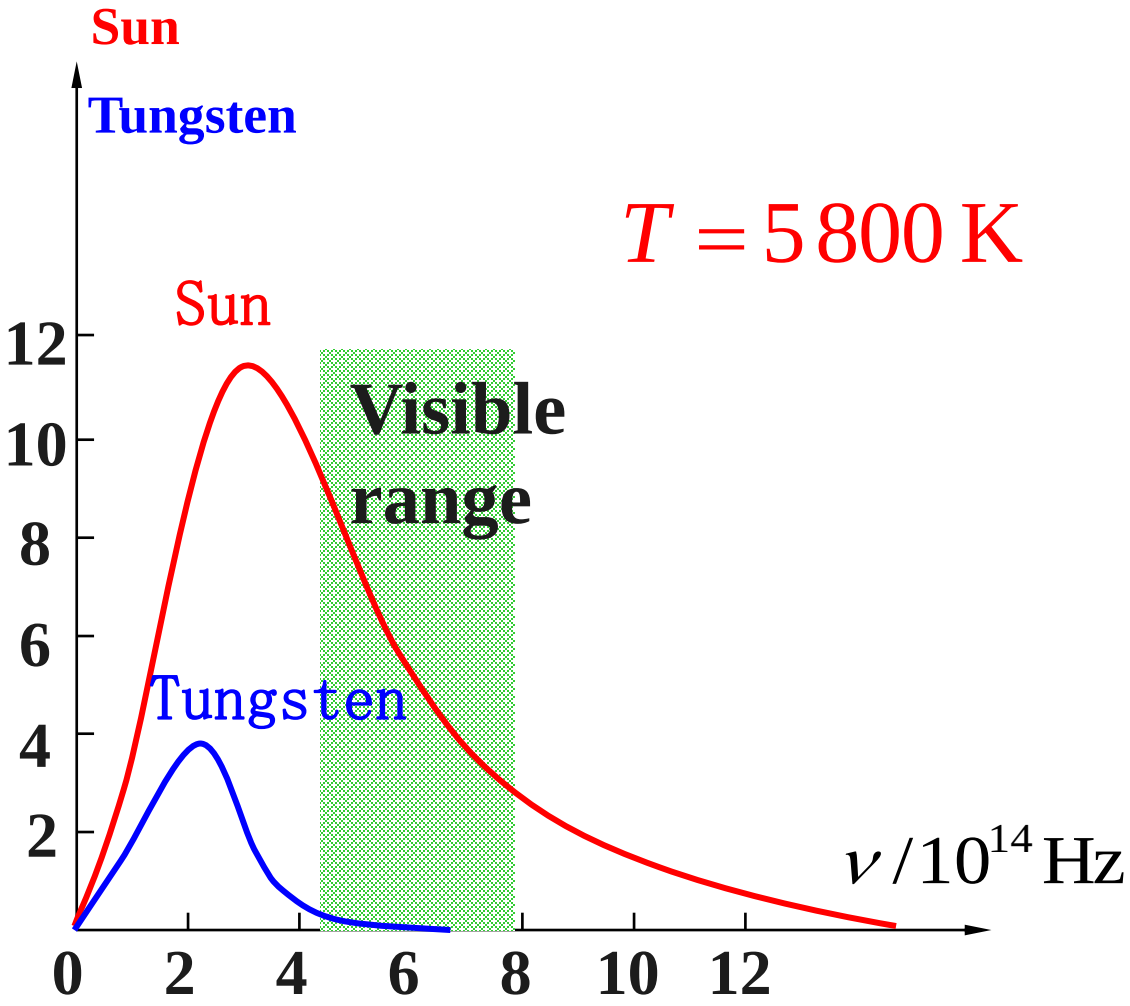
Paschen series $\frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{n^2}\right)$ n=4, 5, …… Infrared

Brackett series $\frac{1}{\lambda} = R\left(\frac{1}{4^2} - \frac{1}{n^2}\right)$ n=5, 6, …… Infrared

Pfund series $\frac{1}{\lambda} = R\left(\frac{1}{5^2} - \frac{1}{n^2}\right)$ n=6, 7, …… Infrared

R: Rydberg constant R=1.097 * 10^7 m^{-1}

11.1 Planck's quantum hypothesis: black body

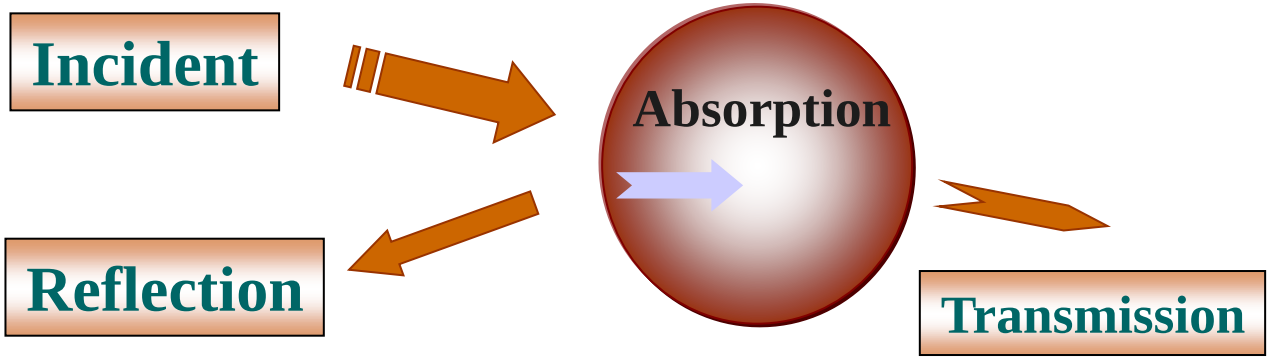


$T = 5\,800$ K

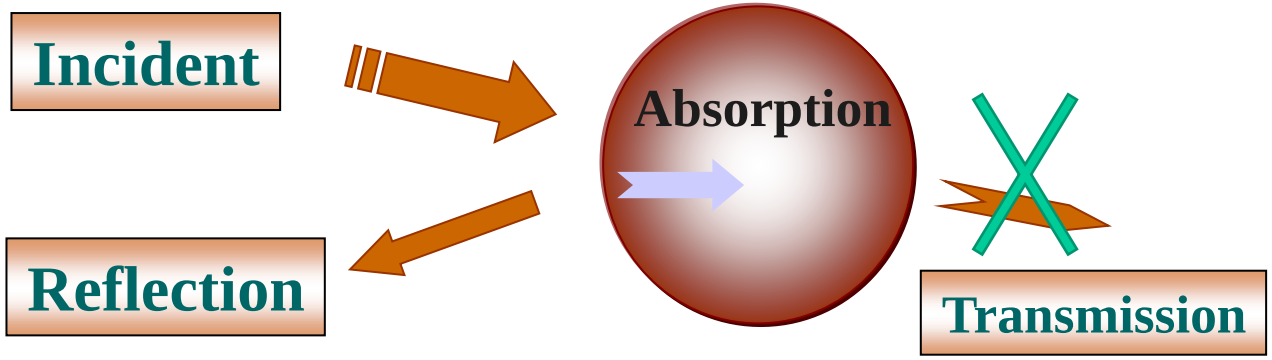
Radiancy

$$R = \int_0^{\infty} R_\lambda \mathrm{d}\lambda$$

11.1 Planck's quantum hypothesis: black body



Not-transparent object

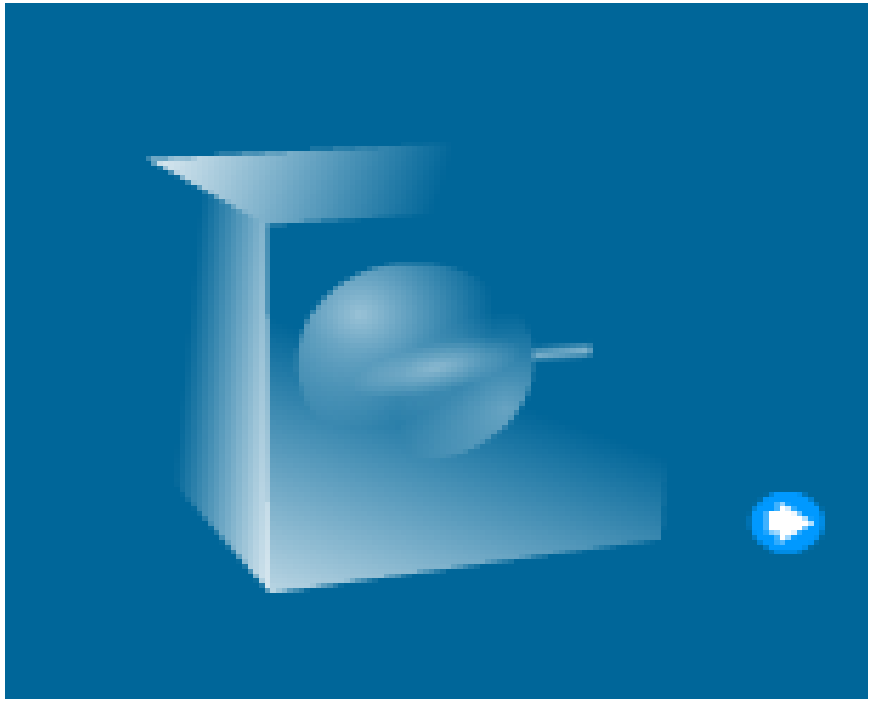


$\alpha_\lambda(T) + r_\lambda(T) = 1$

α_λ.Absorption

R_λ.Reflection

11.1 Planck's quantum hypothesis: black body



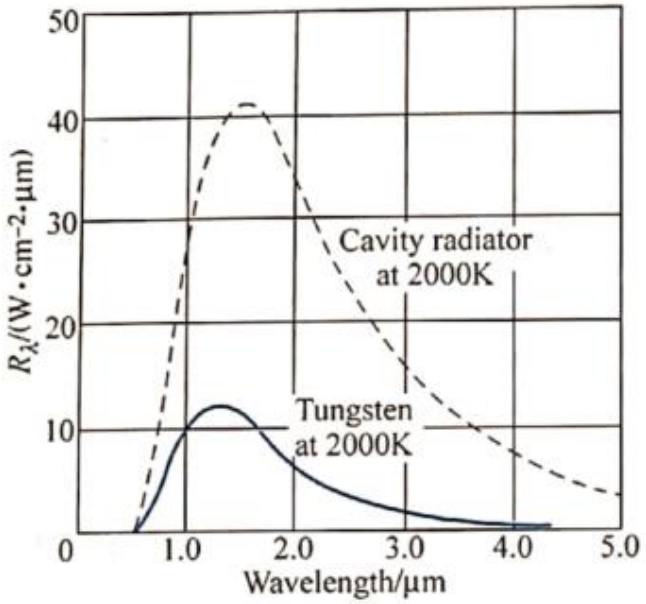
Cavity radiator (black body)

Is an ideal model that can

absorb all incident radiations

(all wavelength radiations).

11.1 Planck's quantum hypothesis: black body

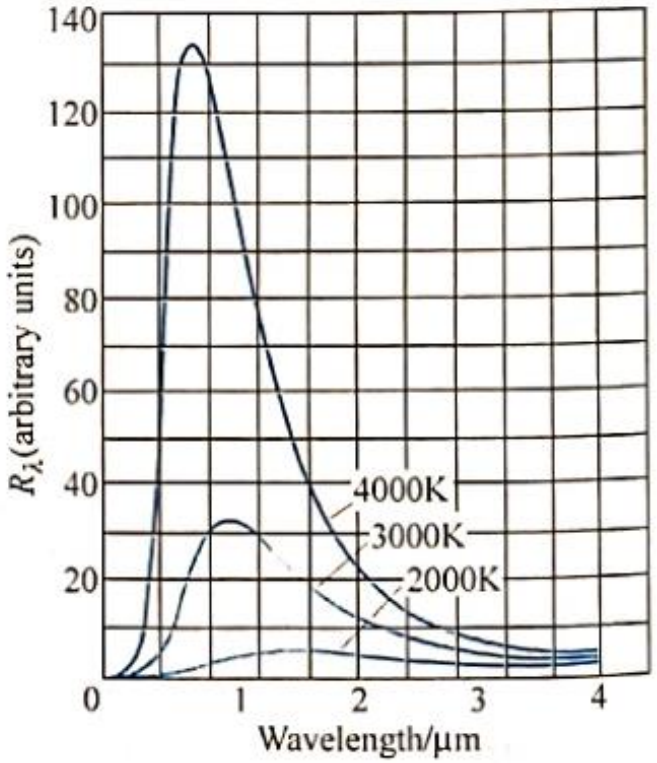


The spectral radiancy of tungsten at 2000K. The dashed curve refers to a cavity radiator at the same temperature.

1. The radiation from the cavity interior is always more intense than the radiation from the outside wall.
2. In contrast to the radiancy of the outer surfaces, the cavity radiancy *Rc* varies with temperature in a simple way:

$$R_\lambda = \sigma T^2$$

11.1 Planck's quantum hypothesis: black body



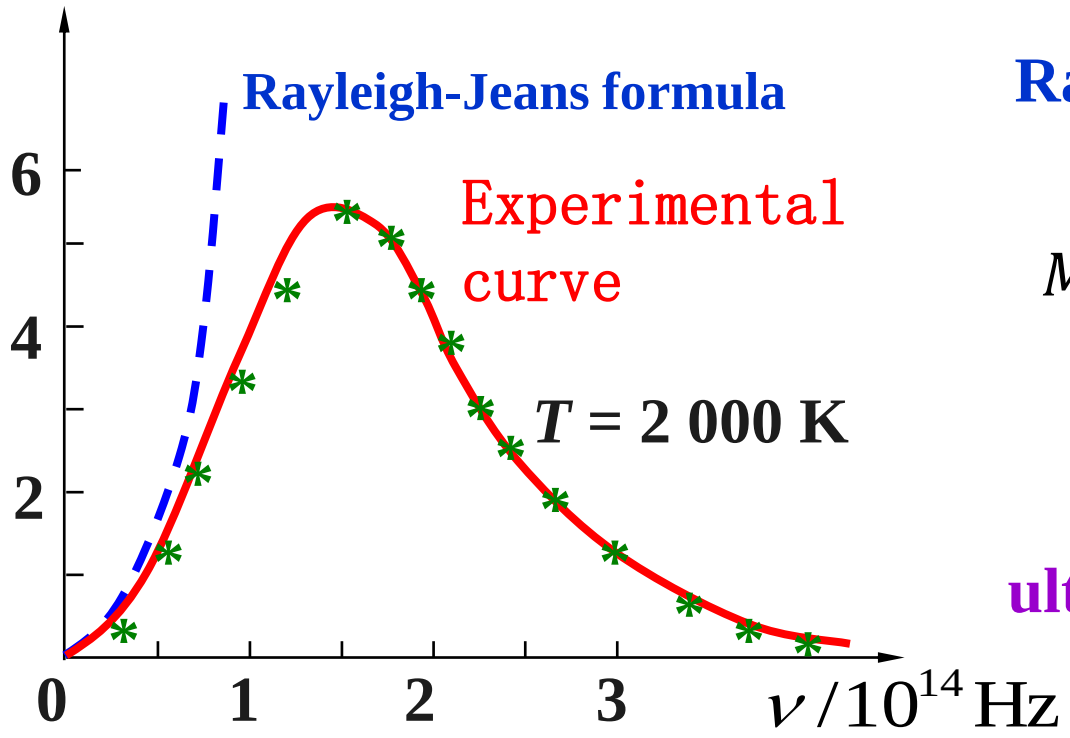
R_λ for the cavity radiation varies with temperature.

These curves depend only on the temperature and are quite independent of the material and of the shape and size of the cavity.

The spectral radiancy for cavity radiation at three different temperatures.

11.1 Black body: Rayleigh-Jeans formula

Classical model: Rayleigh-Jeans formula



Rayleigh-Jeans formula

$$M_\nu(T) = \frac{2\pi\nu^2}{c^2} kT$$

ultraviolet catastrophe

11.1 Planck's quantum hypothesis: black body



Planck（1858 — 1947）

a German theoretical physicist whose discovery of energy quanta won him the Nobel Prize in Physics in 1918.[4]

Planck made many contributions to theoretical physics, but his fame as a physicist rests primarily on his role as the originator of quantum theory,[5] which revolutionized human understanding of atomic and subatomic processes.

https://en.wikipedia.org/wiki/Max_Planck

11.1 Planck's quantum hypothesis: black body

Planck's quantum hypothesis: black body

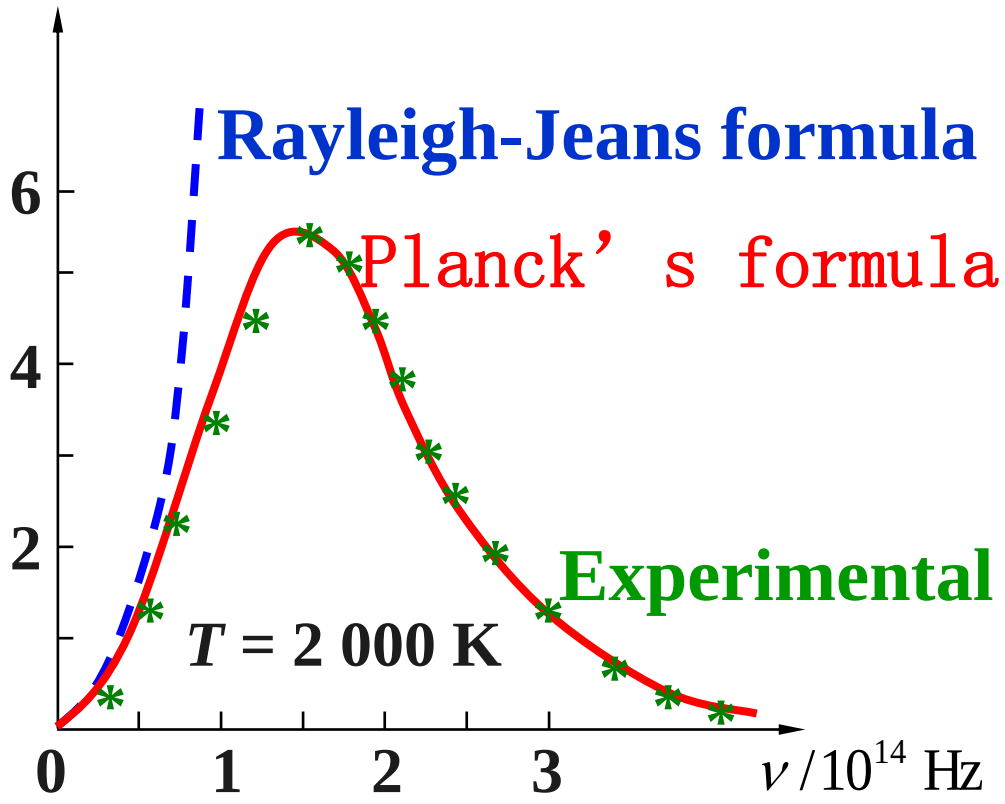
$$R_\lambda = \frac{C_1}{\lambda^5} \frac{1}{e^{c_2/\lambda T} - 1}$$

$$C_1 = 2\pi c^2 h$$

$$C_2 = hc/k$$

1. Assumed that the atoms that make up these walls behave like tiny electromagnetic oscillators, each with a characteristic frequency of oscillation.
2. The oscillators emit electromagnetic energy into the cavity and absorb electromagnetic energy from it.

11.1 Planck's quantum hypothesis: black body



Planck's formula fit the experimental data well!

11.1 Planck's quantum hypothesis: black body

Two radical assumptions:

1. An oscillator cannot have any energy but only energies given

$$E = nh\nu$$ quantized

$$h = 6.626 \times 10^{-34}\,\mathrm{J \cdot s}$$

2. The oscillators do not radiate energy continuously, but only in "jumps", or quanta.

$$\Delta E = \Delta nh\nu = h\nu$$

11.1 Planck's quantum hypothesis: black body

Example 1. A mass-spring system has a mass *m=1.0 Kg* and a spring constant (stiffness) k= 20N m^{-1} and is-oscillating with an amplitude of 1.0cm. (a) If its energy is quantized, what is the quantum number *n?* (b) If *n* changes by unity, what fractional change in energy occurs?

Solution:

(a) The frequency is

$$v = \frac{w}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{20\ N\ m{-}1}{1.0\ kg}} = 0.71\ Hz$$

The mechanical energy of the system is

$$E = \frac{1}{2}kA^2 = \frac{1}{2}(20\ \mathrm{N\ m^{-1}})(1.0{*}10^{-2}\ \mathrm{m})^2 = 1.0 * 10^{-3}\ \mathrm{J}$$

Therefor, the quantum number is

$$n = \frac{E}{h\nu} = \frac{1.0\ *10^{-3}\ J}{(6.63 * 10^{-34}\ J.s)(0.71\ Hz)} = 2.1\ *10^{30}$$

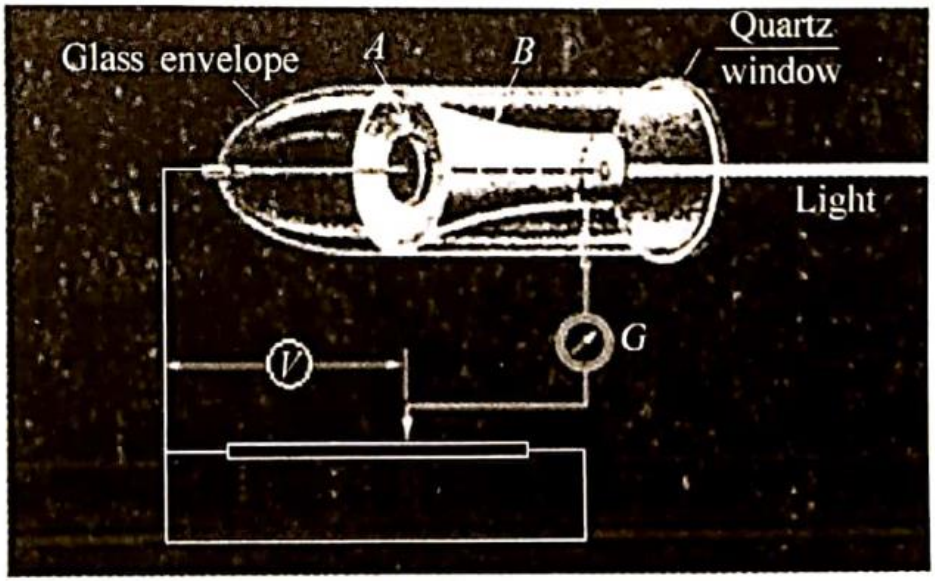
11.1 Planck's quantum hypothesis: black body

(b) The fraction change in energy is:

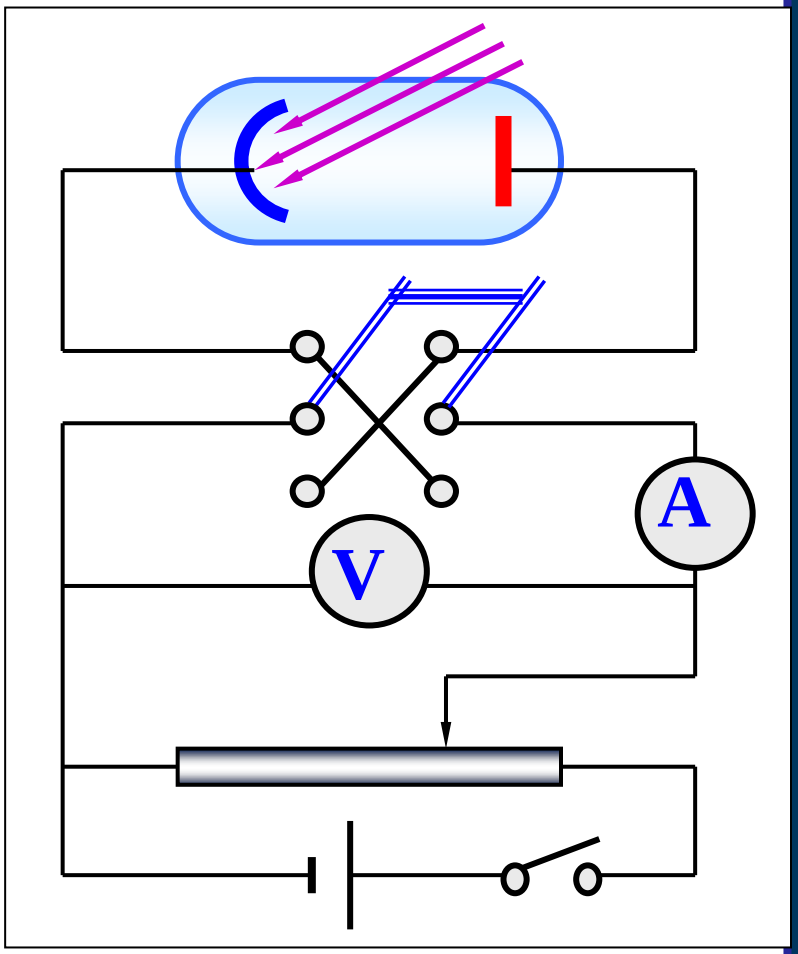
$$\frac{\Delta E}{E} = \frac{h\nu}{nh\nu} = \frac{1}{n} = 4.8 * 10^{-31}$$

Thus for large-scale oscillators the quantum numbers are enormous and the quantized nature of the energy of the oscillations will not be apparent. Similarly, in large-scale experiments Z are not aware of the discrete nature of mass and the quantized nature of charge, that is, of the existence of atoms and electrons.

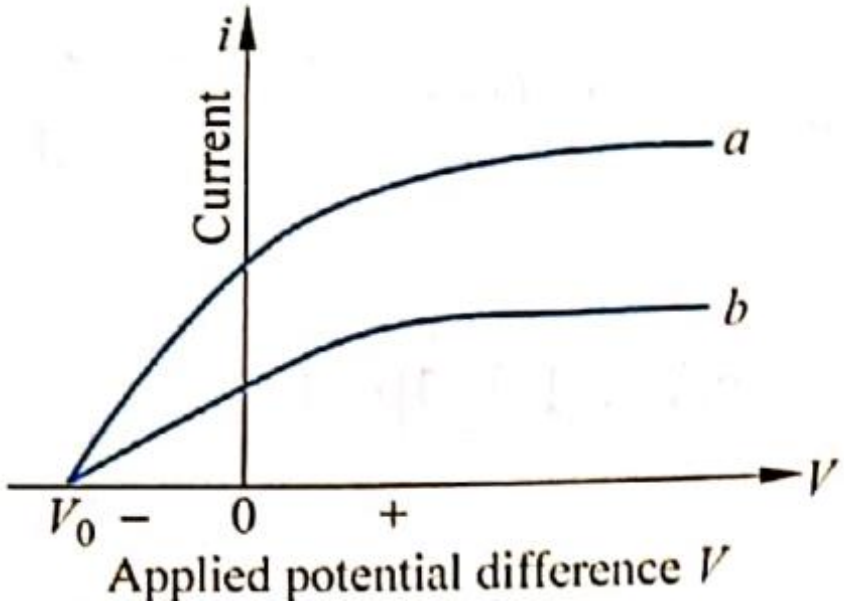
11.2 Photoelectric effect



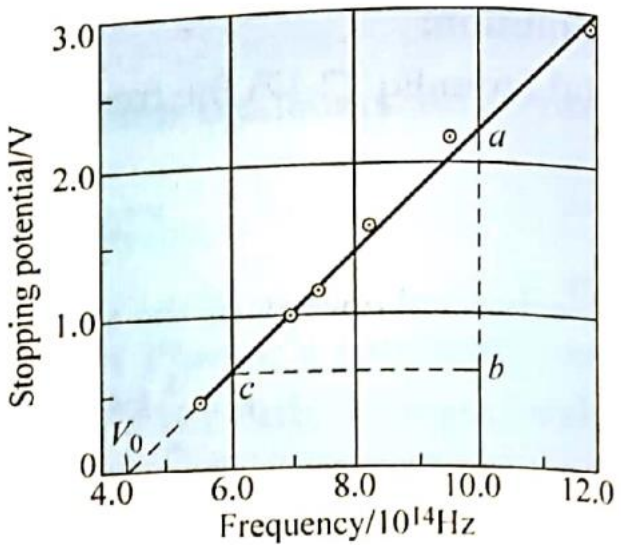
An apparatus used to study the photoelectric effect. V can be varied continuously and can also be reversed in sign by a switching arrangement not shown.



11.2 Photoelectric effect

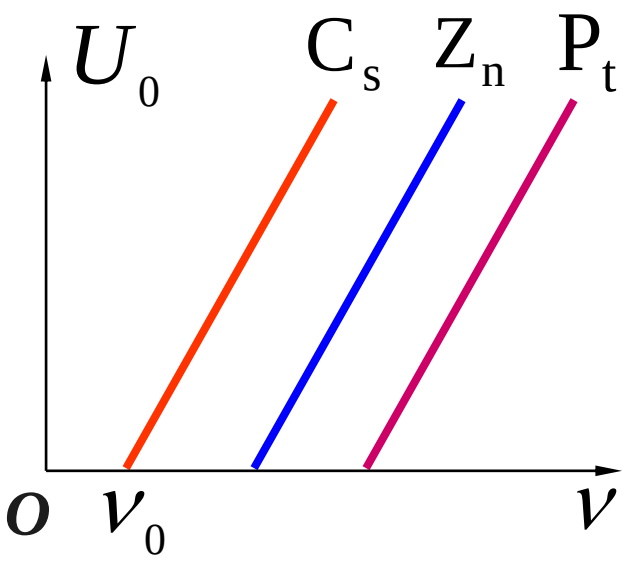


The applied potential difference V is called positive when the cup B is positive with respect to the photoelectric surface A. In curve *b* the incident light intensity has been reduced to one-half that of curve *a*.



A plot of Millikan's measurements of the stopping potential difference at various frequencies for sodium. The cutoff frequency is 4.39 x 10^{14} Hz.

11.2 Photoelectric effect



Three major features of the photoelectric effect cannot be explained by wave theory of light:

1. Wave theory suggests that the kinetic energy of the photoelectrons should increase as the light beam is made more intense. However, Kmax(= eV_0) is independent of the light intensity.

2. According to the wave theory, the photoelectric effect should occur for any frequency of the light, provided only that the light is intense enough. However for each surface, a characteristic cutoff frequency (or threshold frequency)

3. There should be a measurable time lag between the impinging of the light on the surface and the ejection of the photoelectron. However, no detectable time lag has ever been measured.

11.2 Photoelectric effect

$$h\nu = \frac{1}{2}mv^2 + E_0$$

$$h\nu = Kmax + E_0$$

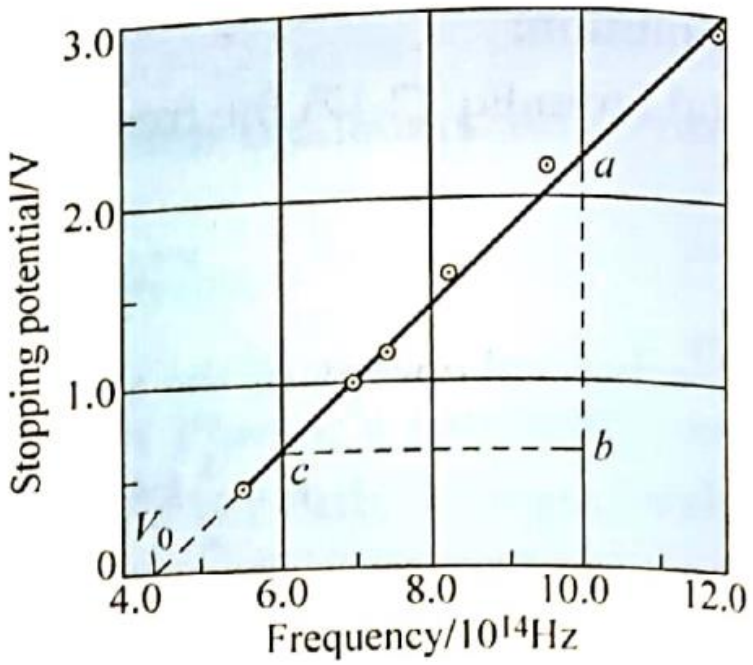
E_0: working function of the materials
Kmax: kinetics of the electrons
or energy of the electrons

E_0: the energy used in causing the electrons to pass through the metal surface

11.2 Work function of some metals

Na	Al	Zn	Cu	Ag	Pt
2.46	4.08	4.31	4.70	4.73	6.35

11.2 Photoelectric effect



$$h\nu = \frac{1}{2}m\mathrm{v}^2 + E_0$$

$$h\nu = qV_0 + E_0$$

$$V_0 = \frac{h}{e}\nu - \frac{E_0}{e}$$

V_0: Stopping potential

$$h = 6.626 \times 10^{-34}\,\mathrm{J}\cdot\mathrm{s}$$

11.2 Photoelectric effect

$$E^2 = (pc)^2 + (mc^2)^2$$

For photon, m=0

$$E = pc$$

$$E = h\nu = hc/\lambda$$

$$p = h/\lambda$$

11.2 Example

Example 1. A metal plate is placed 5m from a monochromatic light source whose power output is 10^{-3} W. Consider that a given ejected photoelectron may collect its energy from a circular area of the plate as large as ten atomic diameters (10^{-9} m) in radius. The energy required to remove an electron through the metal surface is about 5.0eV. Assuming light to be a wave, how long would it take for such a "larger to soak up this much energy from such a light source?

11.2 Example 1

Solution:

The target area is $\pi(10^{-9}\ \mathrm{m})^2$, the area of a 5m sphere centered on the light source is $4\ \pi(5\mathrm{m})^2 = 300\mathrm{m}^2$. Thus, if the light source radiates uniformly in all directions, the rate P at which energy falls on the target is given by

$$P=(10^{-3}\mathrm{W})\left(\frac{3*10^{-18}\mathrm{m}^2}{300\mathrm{m}^2}\right)=10^{-23}\ \mathrm{J\ s}^{-1}$$

The total energy it required

$$Pt=5\ \mathrm{eV}*(1.6*10-19)$$

$$t=5\ \mathrm{eV}*\frac{1.6*10^{-19}}{\mathrm{p}}=22.2\ \mathrm{hours}$$

However, no detectable time lag has been measured

11.2 Example 2

The kinetic energies of photoelectrons range from zero to $4 * 10^{-19}$ J when light of wavelength 300 nm falls on a metal surface,

(a)What is the stopping potential difference for this light?

(b) What is the threshold wavelength for this metal?

11.2 Example 2

Solution:

(a) From the problem we know that

$$K_{max} = 4.0 * 10^{-19}\,J$$

the stopping potential difference

$$V_0 = \frac{Kmax}{e} = \frac{\mathbf{4.0*10^{-19}\,J}}{1.6 * 10^{-19}\,C} = 2.5\,V$$

11.2 Example 2

(b) For the threshold wavelength for this metal:

$$h\nu = K_{max} + E_0$$

$$hc/\lambda_0 = E_0 = hc/\lambda - Kmax$$

$$1/\lambda_0 = 1/\lambda - Kmax\ /\mathbf{hc}$$

λ_0**=765 nm**

Homework

Page 279， Q1, 3, 6,7， 8

Submit before next Monday’s class

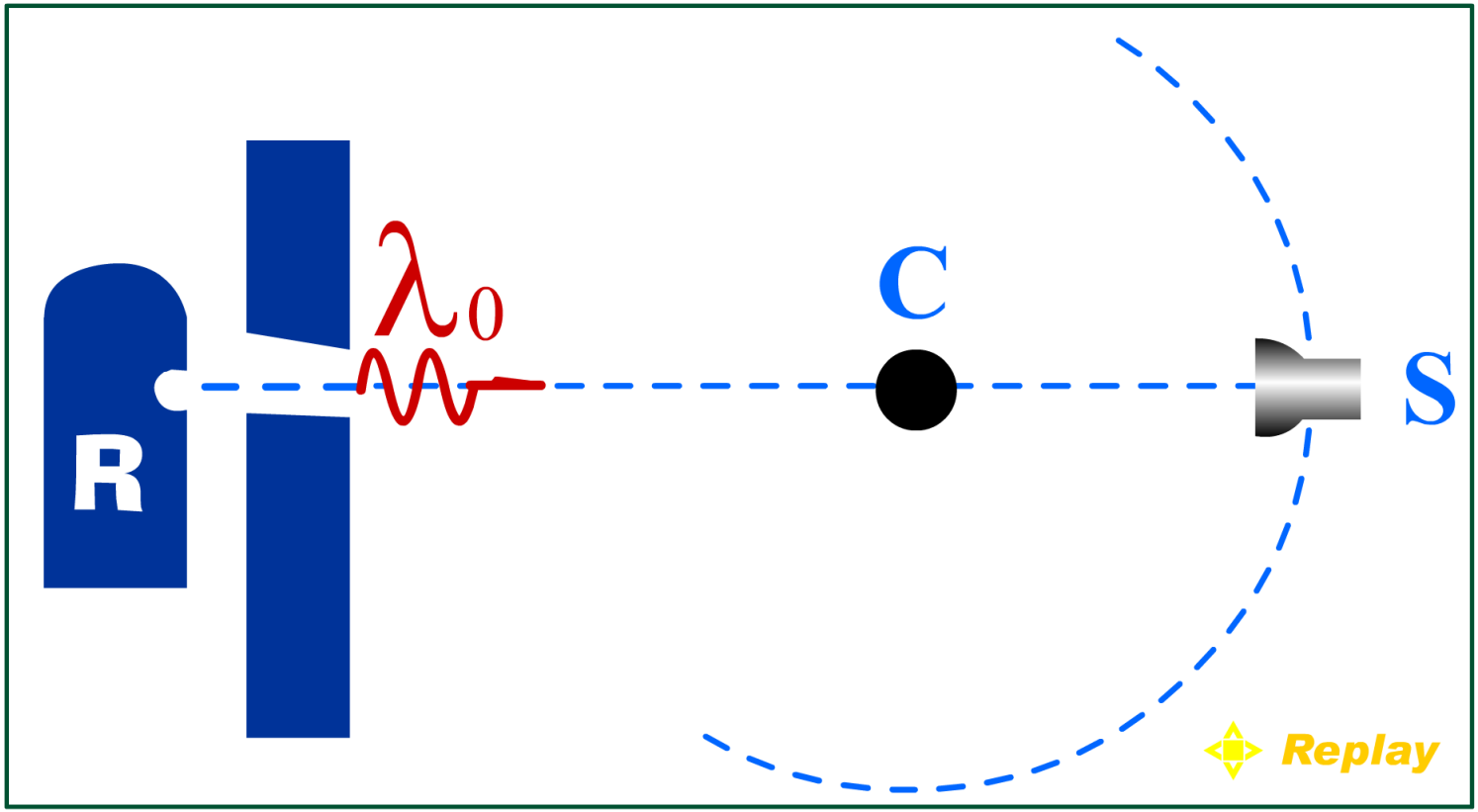
Homework file name:

Roll number+week 13 + name,

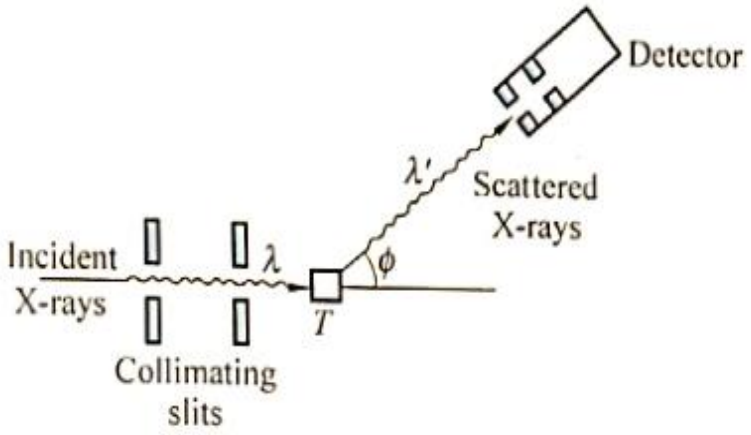
for example: 2023800901XX + week 13 + Alfa Robot

pdf file, send to the cloud)

11.2 The Compton effect



11.2 The Compton effect



A beam of X-rays falls on a graphite target T. The X-rays scattered from the target are observed at various angles to the incident direction.

X-rays striking a target are scattered at various angles by electrons in the target. A shift in wavelength is observed for the scattered X-rays , and the phenomenon is known as the Compton effect.

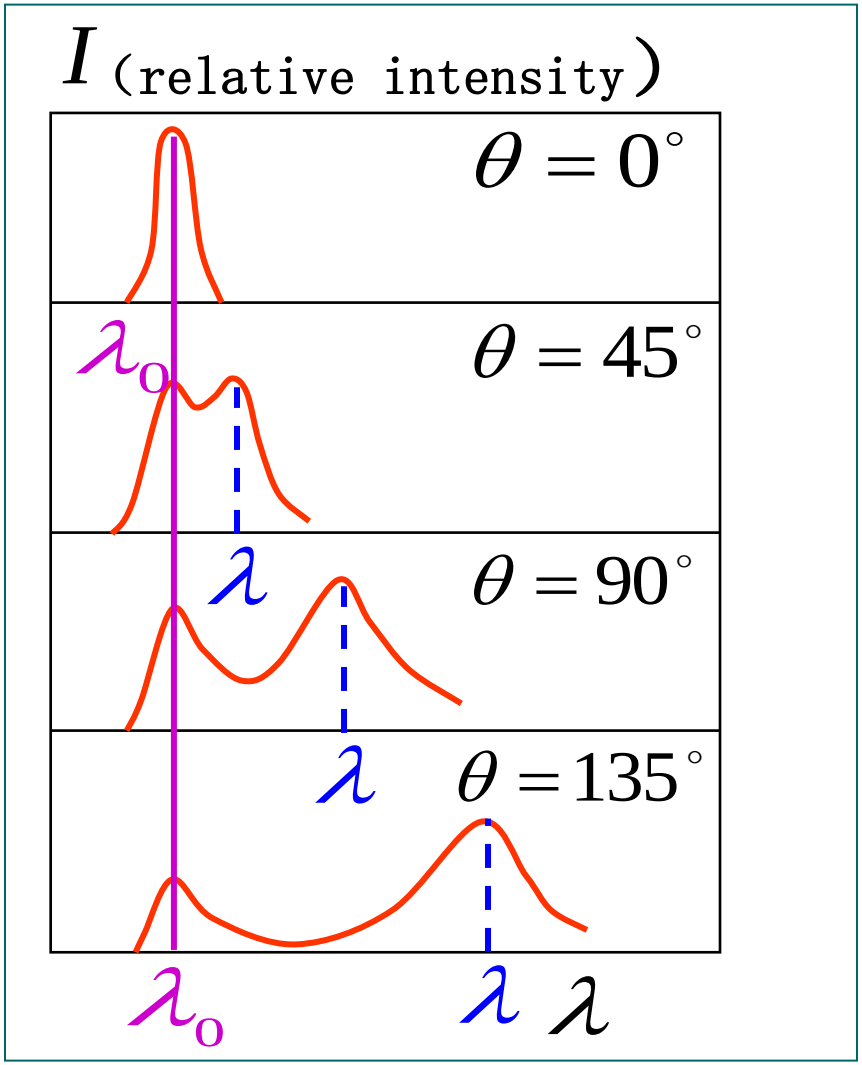
Classical physics does not correctly explain the experimental results of this effect.

11.2 The Compton effect

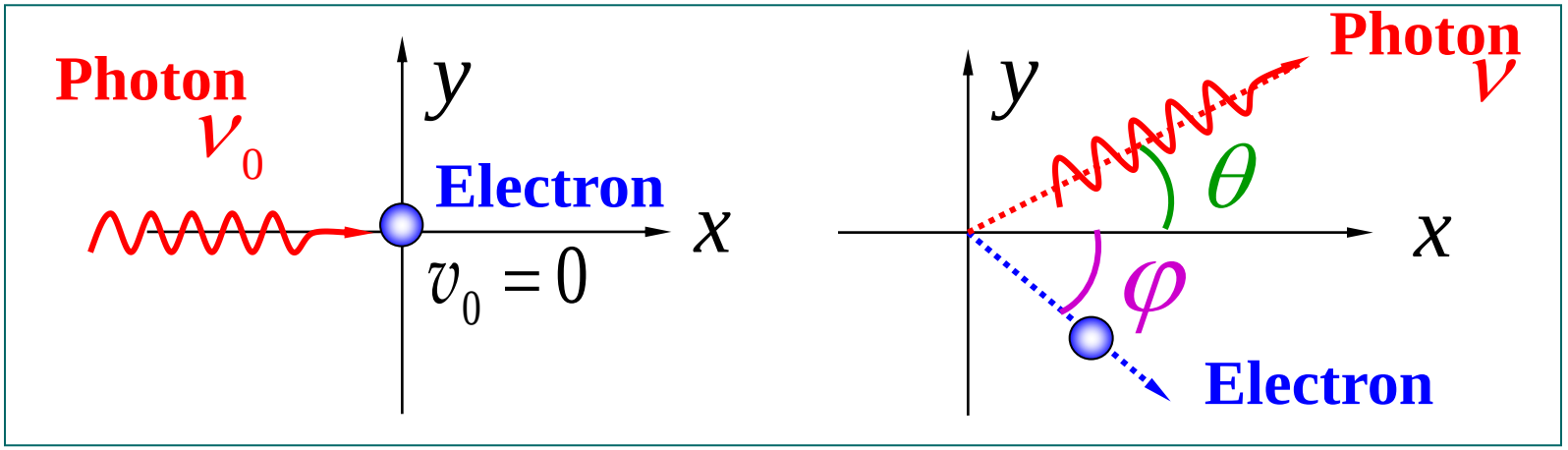
1. Wavelength shift is relative to the scatter angle

$$\Delta\lambda = \lambda - \lambda_0$$

2. $\Delta\lambda$ is not related to the object



11.2 The Compton effect: physical model

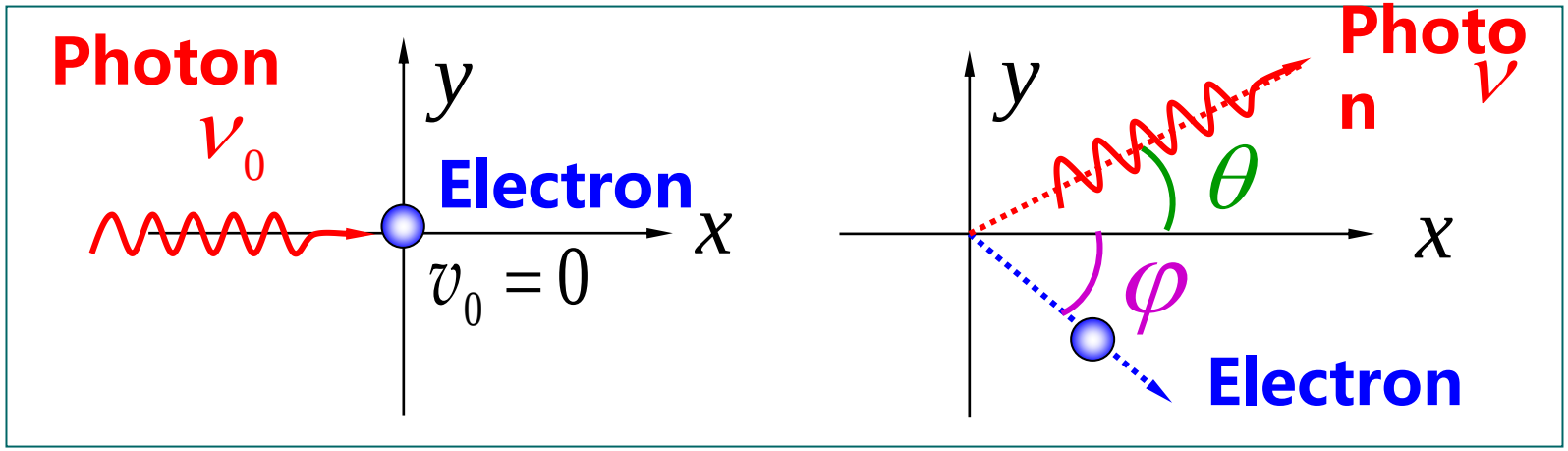


Incident x-ray or γ-ray with a high energy

$$E = h\nu \quad : \qquad 10^4 \sim 10^5 \text{eV}$$

11.2 The Compton effect

The electron is treated as stable or v=0



- The binding energy of the electron is so weak that it can be treated as free electrons
- The speed of electrons after scattering is so fast that we need to use the Relativity theory.

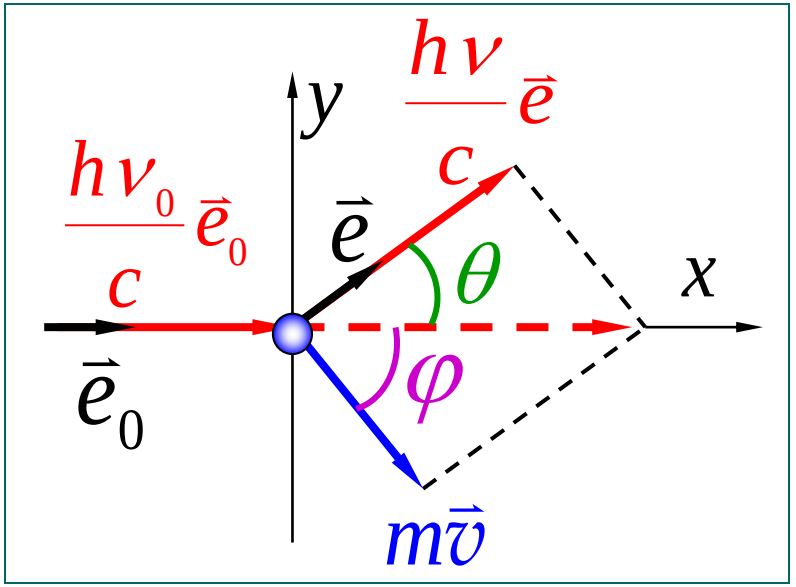
11.2 The Compton effect

Energy conservation

$$h\nu_0 + m_0c^2 = h\nu + mc^2$$

Momentum conservation

$$\frac{h\nu_0}{c}\vec{e}_0 = \frac{h\nu}{c}\vec{e} + m\vec{v}$$



$$m^2v^2 = \frac{h^2\nu_0^2}{c^2} + \frac{h^2\nu^2}{c^2} - 2\frac{h^2\nu_0\nu}{c^2}\cos\theta$$

11.2 The Compton effect

$$m^2 v^2 = \frac{h^2 \nu_0^2}{c^2} + \frac{h^2 \nu^2}{c^2} - 2\frac{h^2 \nu_0 \nu}{c^2}\cos\theta$$

$$m^2 c^4 (1 - \frac{v^2}{c^2}) = m_0^2 c^4 - 2h^2 \nu_0 \nu (1 - \cos\theta) + 2m_0 c^2 h(\nu_0 - \nu)$$

$$m = m_0 (1 - v^2 / c^2)^{-1/2}$$

$$\frac{c}{\nu} - \frac{c}{\nu_0} = \frac{h}{m_0 c}(1 - \cos\theta) \ = \lambda - \lambda_0 = \Delta\lambda$$

11.2 The Compton effect

$$\Delta\lambda = \frac{h}{m_0 c}(1-\cos\theta) = \frac{2h}{m_0 c}\sin^2\frac{\theta}{2}$$

- **Compton wavelength** $\lambda_C = \dfrac{h}{m_0 c} = 2.43\times10^{-12}\ \mathrm{m}$

- **Compton shift**

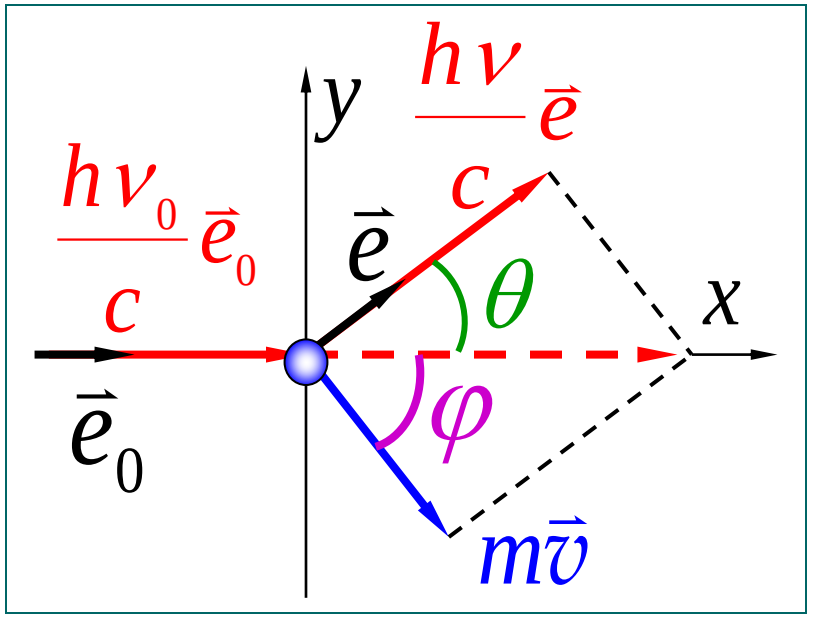
$$\Delta\lambda = \frac{h}{m_0 c}(1-\cos\theta) = \lambda_C(1-\cos\theta)$$

11.2 The Compton effect: conclusion

Scatter wavelength change Δλ is only related to scatter angle θ

$$\theta = 0, \Delta\lambda = 0$$

$$\theta = \pi, \quad (\Delta\lambda)_{\max} = 2\lambda_C$$



The energy of the scattered ray is reduced

$$\lambda > \lambda_0, \nu < \nu_0$$

11.2 The Compton effect: further discussion

- Photons have wave-particle nature

Generally, light/photons exhibit wave nature when transmit.

Light/photons exhibit particle nature when interact with subject (electrons, for example)

- If $\lambda_0 >> \lambda_C$, $\lambda \approx \lambda_0$ we can not see the Compton effect for the visible light

11.2 The Compton effect: conclusion

- Demonstrate that the photon is particle nature, and Relativity theory is correct.

- The high energy particles collision are also follow the energy and momentum conservation.

11.2 Example

X-rays with λ =1.00 Å(=0.1 nm) are scattered from a carbon block. The scattered radiation is viewed at 90° to the incident beam, (a) What is the Compton shift $\Delta\lambda$? (b) What kinetic energy is imparted to the recoiling electron?

11.2.1 Example

Solution:

(a) The Compton shift:

$$\Delta\lambda = \frac{h}{m_o c}(1-\cos\theta) = \frac{6.63 * 10^{-34}\ \text{J s}}{(9.11*10^{-31}\ \text{Kg})(3*108\ \text{m s}^{-1})}(1\text{-}\cos 90^{o}) = 0.0243\ \text{Å}$$

(b) If K is the kinetic energy of the electron

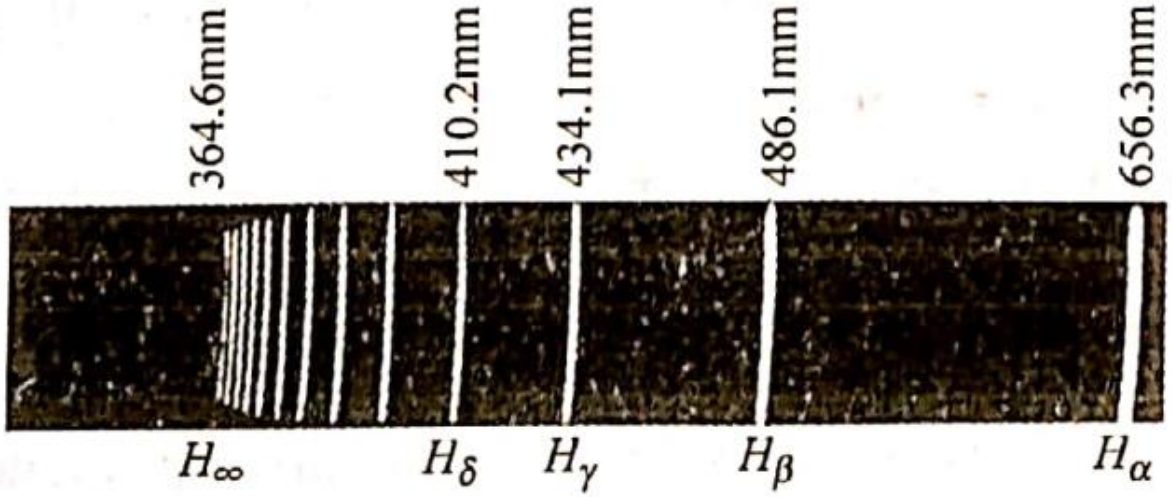
$$h\nu = h\nu' + K$$

$$hc/\lambda = hc/\lambda' + K$$

$$K = \frac{hc\Delta\lambda}{\lambda(\lambda+\Delta\lambda)} = \frac{(6.63 * 10^{-34}\ \text{Js})(3.00*10^{8}\ \text{m s}^{-1})(2.43 * 10^{-12}\ \text{m})}{(1.00*10^{-10}\ \text{m}))(1.00+0.024)1.00*10^{-10}\ \text{m}}$$

$= 4.73 * 10^{-17}$ J = 295 eV

11.2.2 The Bohr hydrogen atom



The Balmer series of atomic hydrogen

A line spectrum is a series of discrete electromagnetic wavelengths emitted by the atoms of a low pressure gas that is subjected to a sufficiently-high potential difference.

Certain groups of discrete wavelengths are referred to as “series”. The line spectrum of atomic hydrogen includes, among others, the Lyman, Balmer, Paschen, Brackett, Pfund series.

11.2.2 The Bohr hydrogen atom

Lyman series	$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$	n=2, 3, …… Ultraviolet
Balmer series	$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$	n=3, 4, …… Visible
Paschen series	$\frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{n^2}\right)$	n=4, 5, …… Infrared
Brackett series	$\frac{1}{\lambda} = R\left(\frac{1}{4^2} - \frac{1}{n^2}\right)$	n=5, 6, …… Infrared
Pfund series	$\frac{1}{\lambda} = R\left(\frac{1}{5^2} - \frac{1}{n^2}\right)$	n=6, 7, …… Infrared

R: Rydberg constant R=1.097 * 10^7 m^{-1}

11.2.2 The Bohr hydrogen atom

For an electron in the electric field by a proton

$$\frac{e^2}{4\pi\varepsilon_0 r^2} = m\frac{v^2}{r}$$

The kinetic energy of the electron is:

$$K = \frac{1}{2}mv^2 = \frac{e^2}{8\pi\varepsilon_0 r^2}$$

The potential of the proton at the radius of the electrons:

$$V = \frac{e}{4\pi\varepsilon_0 r}$$

Therefore, the total energy E of the system is:

$$E = K + U = \frac{e^2}{8\pi\varepsilon_0 r^2} + \frac{-e^2}{4\pi\varepsilon_0 r}$$

11.2.2 The Bohr hydrogen atom

Since the orbit radius can apparently take on any value, so can the energy E. The problem of quantizing E reduces to that of quantizing r.
Every property of the orbit is fixed if the radius is given.

The linear momentum p is :

$$p = mv = \sqrt{\frac{me^2}{4\pi\varepsilon_0 r}}$$

The angular momentum L is :

$$L = pr = \sqrt{\frac{me^2 r}{4\pi\varepsilon_0}}$$

Thus if r Is known, the orbit parameters *K U*, *E*, *v*, v_0, *p* and *L* are also known. If any one of these quantities is quantized, all of them must be.

11.2.2 The Bohr hydrogen atom

Bohr has a bold hypothesis: quantization of the orbit parameters, which means L can take on only values given by

$$L = n\frac{h}{2\pi} \qquad n=1, 2, 3, \ldots\ldots$$

$$r=n^2\frac{h^2\varepsilon_0}{\pi m e^2}$$

$$E=-\frac{me^4}{8\varepsilon_0{}^2h^2}\frac{1}{n^2} \qquad E=-\frac{13.6\ eV}{n^2}$$

$$h\nu=\frac{hc}{\lambda}=\frac{me^4}{8\varepsilon_0{}^2h^2}\left(\frac{1}{n_f{}^2}-\frac{1}{n_i{}^2}\right)$$

$$\frac{1}{\lambda}=R\left(\frac{1}{n_f{}^2}-\frac{1}{n_i{}^2}\right)$$

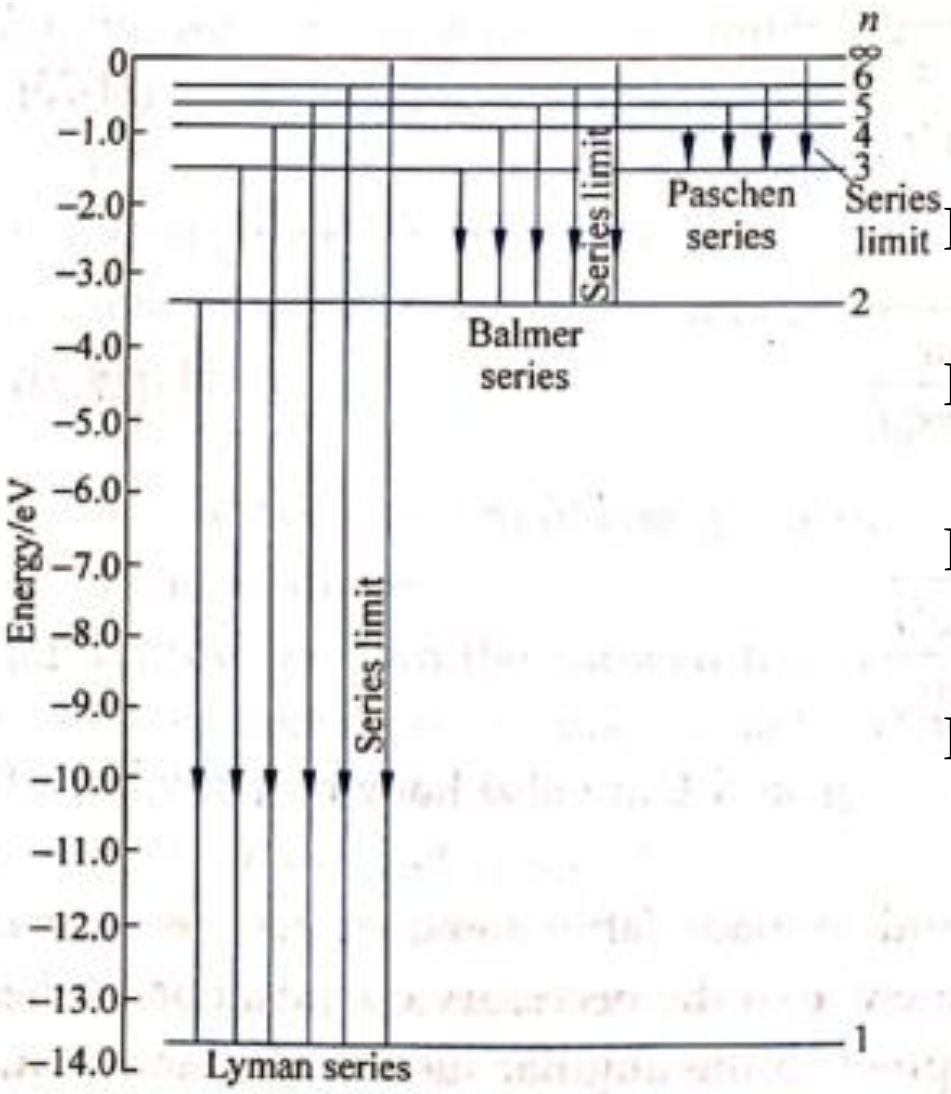
11.2.2 The Bohr hydrogen atom

Here n_i, and n_f are the quantum numbers of the upper and the lower energy states involved in the transition whose wavelength is λ.

The factor *R*, called the Rydberg constant:

$$R=\frac{me^4}{8\varepsilon_0{}^2h^3c}=1.097*10^7\ m^{-1}$$

11.2.2 The energy level diagram for hydrogen



Lyman series $\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n^2}\right)$

Balmer series $\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$

Paschen series $\frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{n^2}\right)$

Brackett series $\frac{1}{\lambda} = R\left(\frac{1}{4^2} - \frac{1}{n^2}\right)$

Pfund series $\frac{1}{\lambda} = R\left(\frac{1}{5^2} - \frac{1}{n^2}\right)$

11.2.2 The Bohr hydrogen atom

(a) What is the wavelength of the least energetic photon in the Balmer series?
(b) What is the wavelength of the series limit for the Balmer series?

11.2.2 The Bohr hydrogen atom

Solution:

(a) For the Balmer series: $n_i=3$ and $n_f=2$. From the relation $E=h\nu$, the least energetic photon has the smallest frequency and thus the greatest wavelength.

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f{}^2} - \frac{1}{n_i{}^2}\right) = 1.097 * 10^7\ \text{m}^{-1}\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 1.524 * 10^6\ \text{m}^{-1}$$

$$\lambda = 656.3\ nm$$

(b) For the Balmer series: $n_f=2$. The series limit, then $n_i=\infty$

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f{}^2} - \frac{1}{n_i{}^2}\right) = 1.097 * 10^7\ \text{m}^{-1}\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = 2.743 * 10^6\ \text{m}^{-1}$$

$$\lambda = 364.6\ nm$$

11.2.2 The Bohr hydrogen atom

Example 2. Calculate the binding energy of the hydrogen atom (the energy binding the electron to the nucleus).

Solution:
The binding energy is numerically equal to the energy of the lowest state.

$$E=-\frac{13.6\ eV}{n^2}=-\frac{13.6\ eV}{1}=-13.6\ eV$$

Bohr model

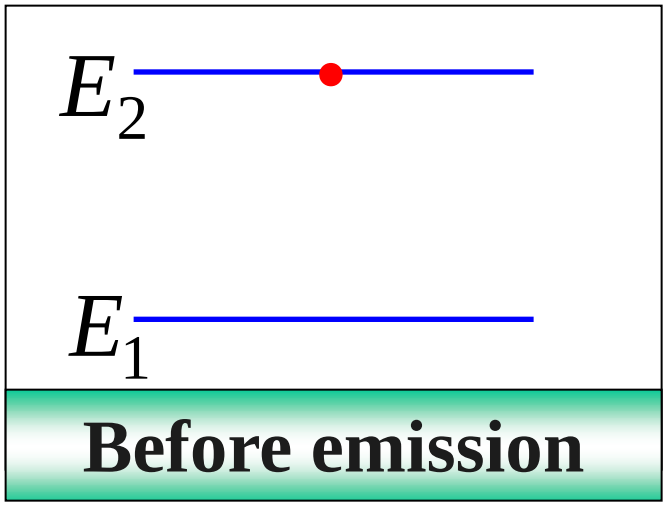
The Bohr model assumes that the electron exists in circular orbits that are called stationary orbits because the electron does not radiate electromagnetic waves while in them. According to this model, a photon is emitted only when an electron changes from a higher energy orbit to a orbital angular momentum L of the electron can only have values that are integer multiplies of Planck's constant divided by 2π

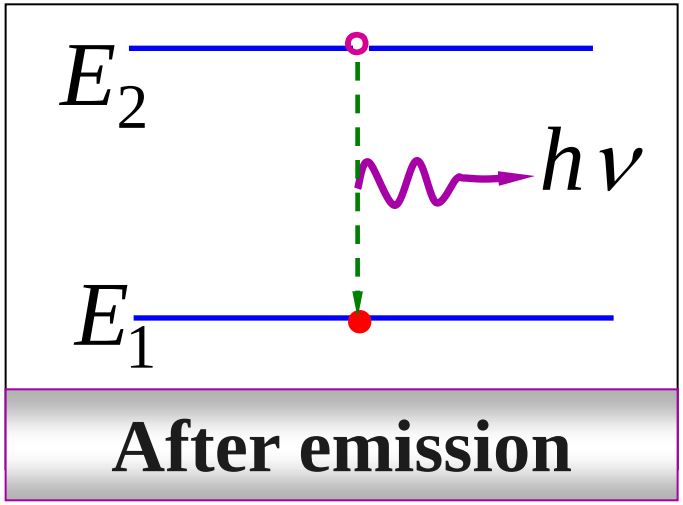
$$L = n\frac{h}{2\pi}$$ **n=1, 2, 3, ……**

11.2.3 Laser

Laser: an acronym for light amplification by the stimulated emission of radiation.

Self-radiation

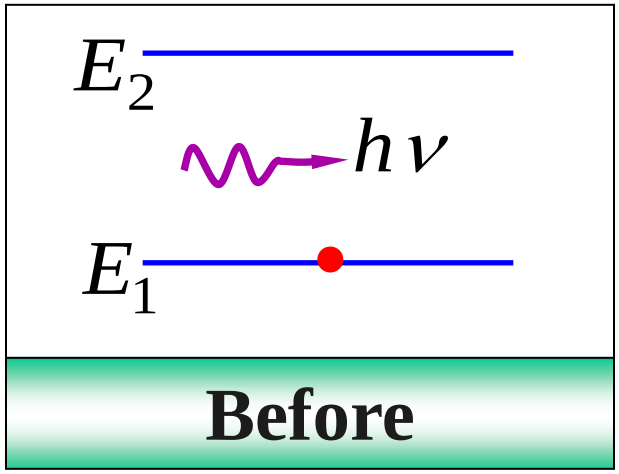


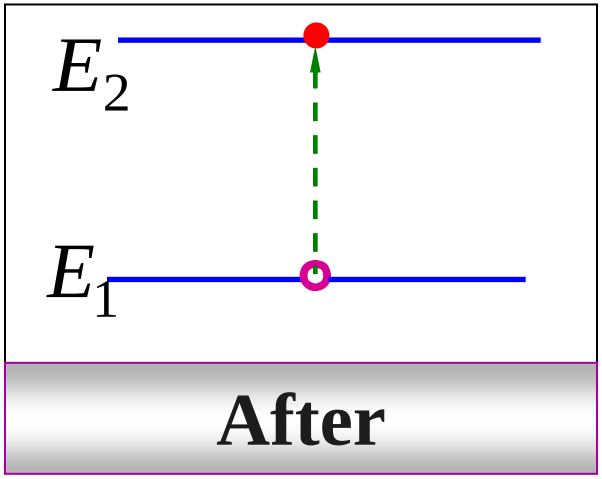


$$\nu = \frac{E_2 - E_1}{h}$$

11.2.3 laser: absorption

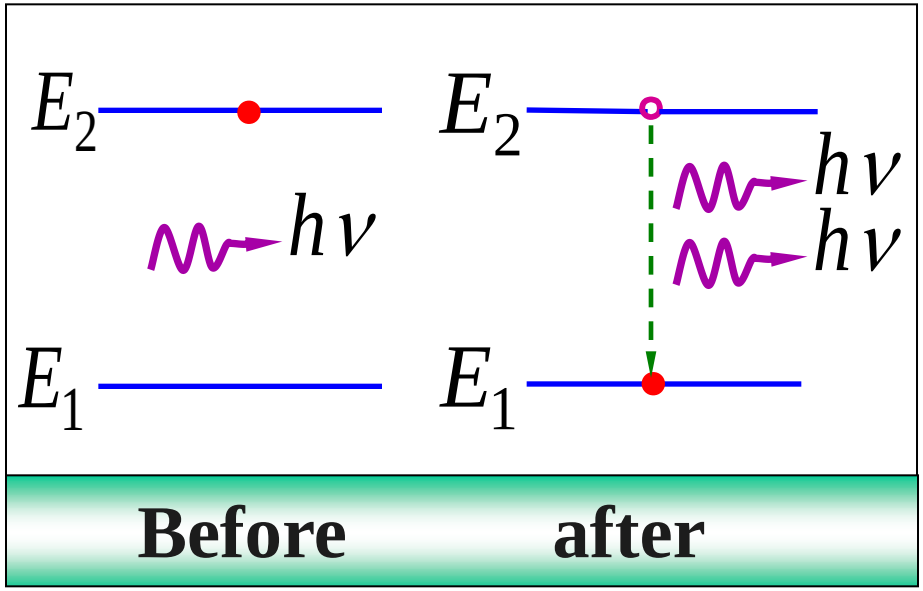
Atom absorb the energy from photons, electrons jump from low energy to high energy

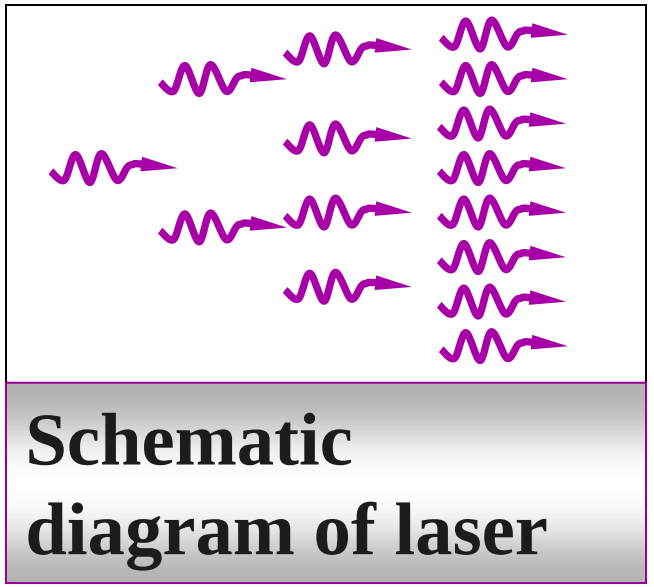




11.2.3 laser: stimulated emission of radiation

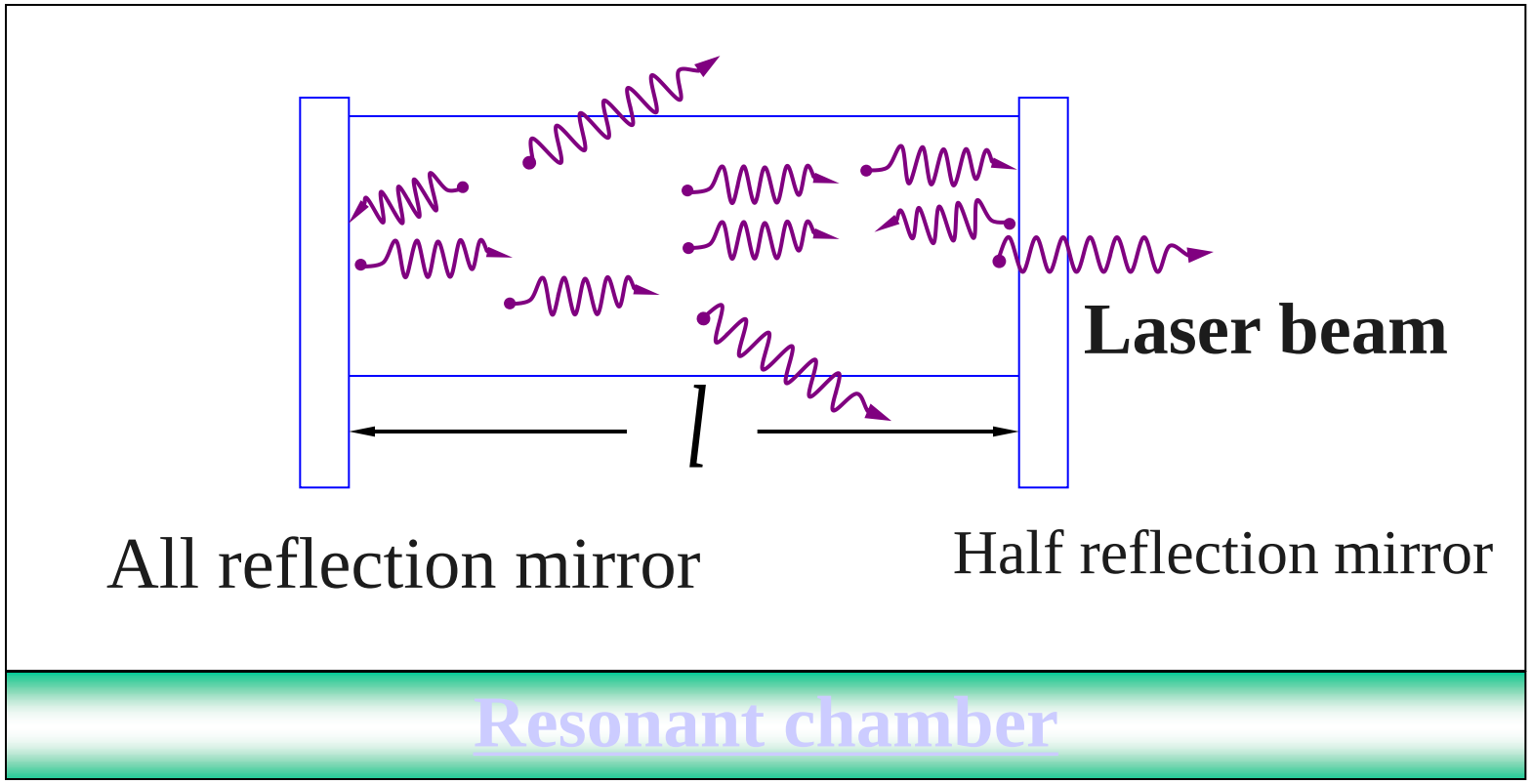
Stimulated emission of radiation





Schematic diagram of laser

11.2.3 laser: stimulated emission of radiation



11.3 The wave nature of matter

Light:

Particle-like

$$\begin{cases} E = mc^2 = h\nu \\ P = mv = h/\lambda \end{cases}$$

Wave-like

How about the other matters?

11.3 The wave nature of matter

De Broglie（1892 — 1987）



A French physicist who made groundbreaking contributions to quantum theory. In his 1924 PhD thesis, he postulated the wave nature of electrons and suggested that all matter has wave properties. This concept is known as the de Broglie hypothesis, an example of wave–particle duality, and forms a central part of the theory of quantum mechanics.

11.3 The wave nature of matter

Wavelength of matter wave

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mk}}$$

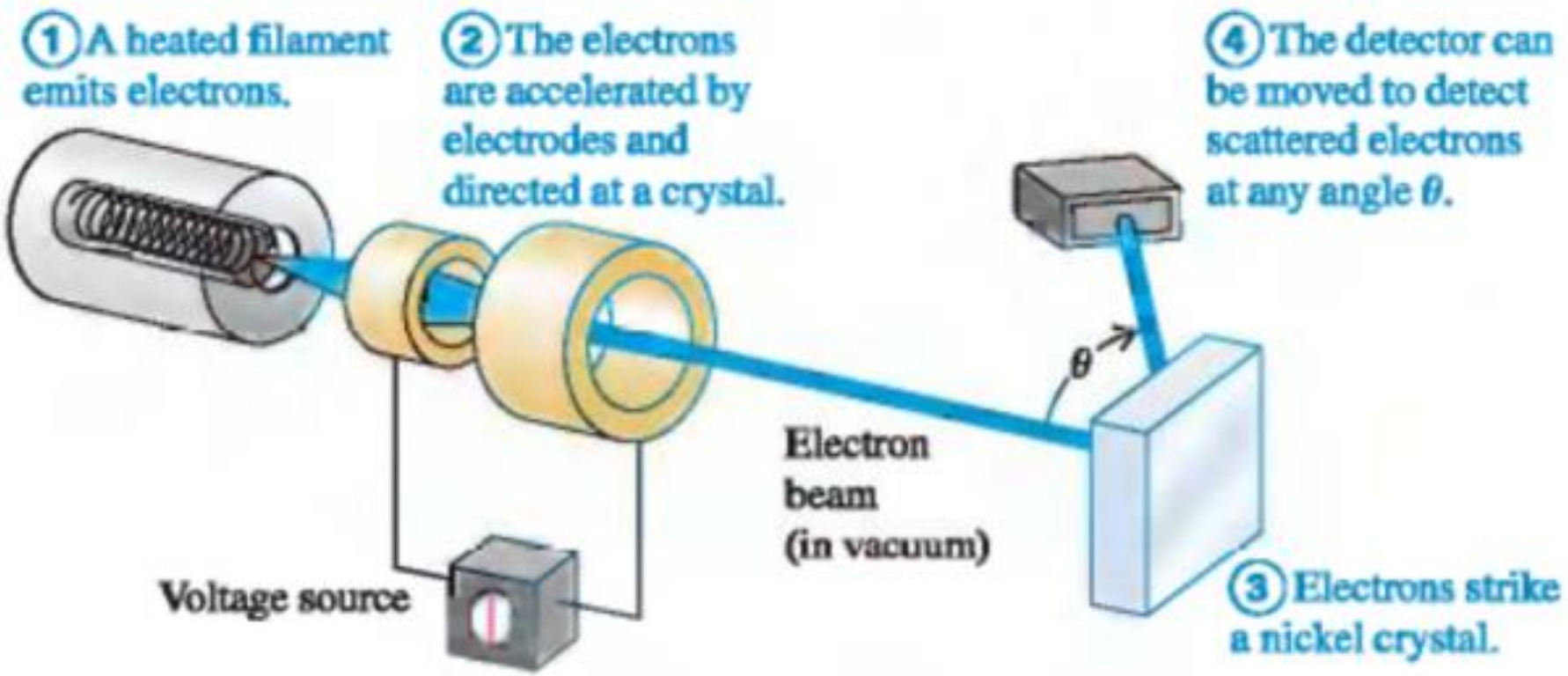
The kinetic energy:

$$K = \frac{p^2}{2m}$$

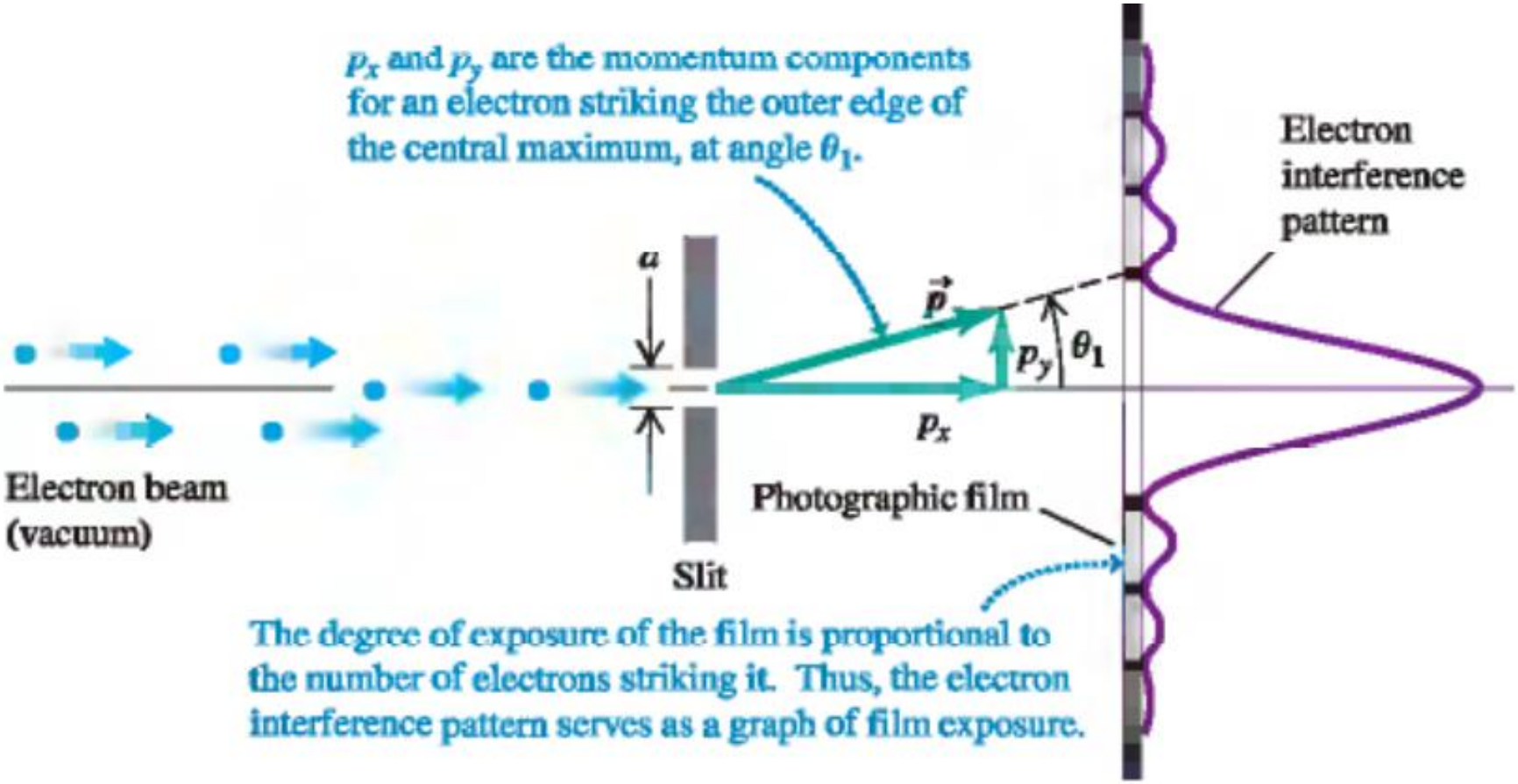
For an electron with the kinetic energy of 54 eV, the calculated de Broglie wavelength:

$$\lambda = \frac{6.63 * 10^{-34}\ \text{J s}}{\sqrt{2(9.11 * 10^{-31}\ \text{kg})(54 * 1.6 * 10-19\ \text{J})}} = 1.67 * 10^{-10}\ \text{m}$$

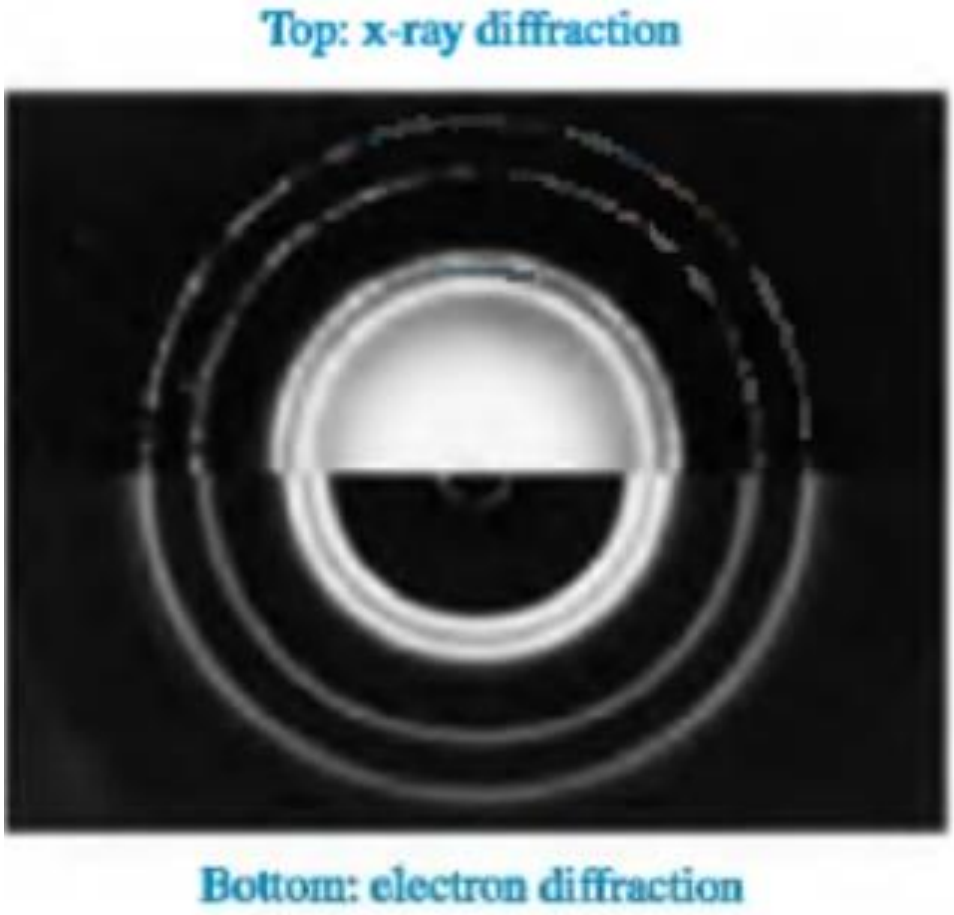
11.3.2 The electron microscopy-TEM



11.3.2 An electron diffraction experiment

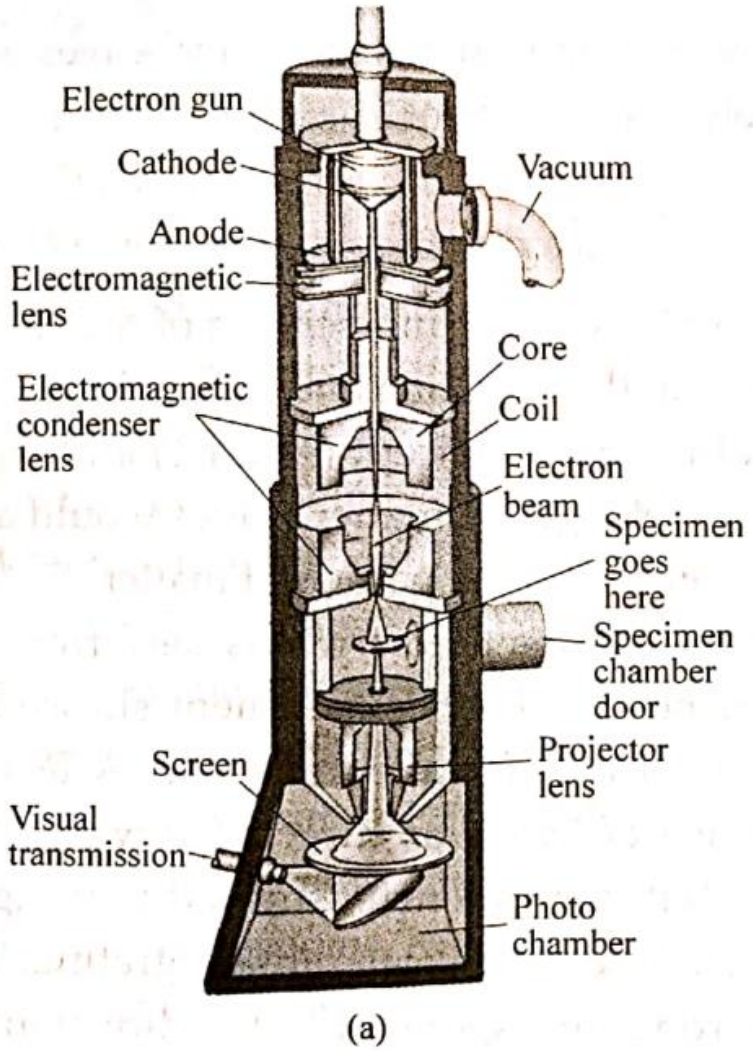


11.3 The wave nature of matter



X-ray and election diffraction. The upper half of the photo shows the diffraction pattern for 71-pm x-rays passing through Al foil. The lower half, with different scale, shows the diffraction pattern for 600 eV electrons from Al.

11.3.2 The electron microscopy-TEM





11.3.2 The electron microscopy

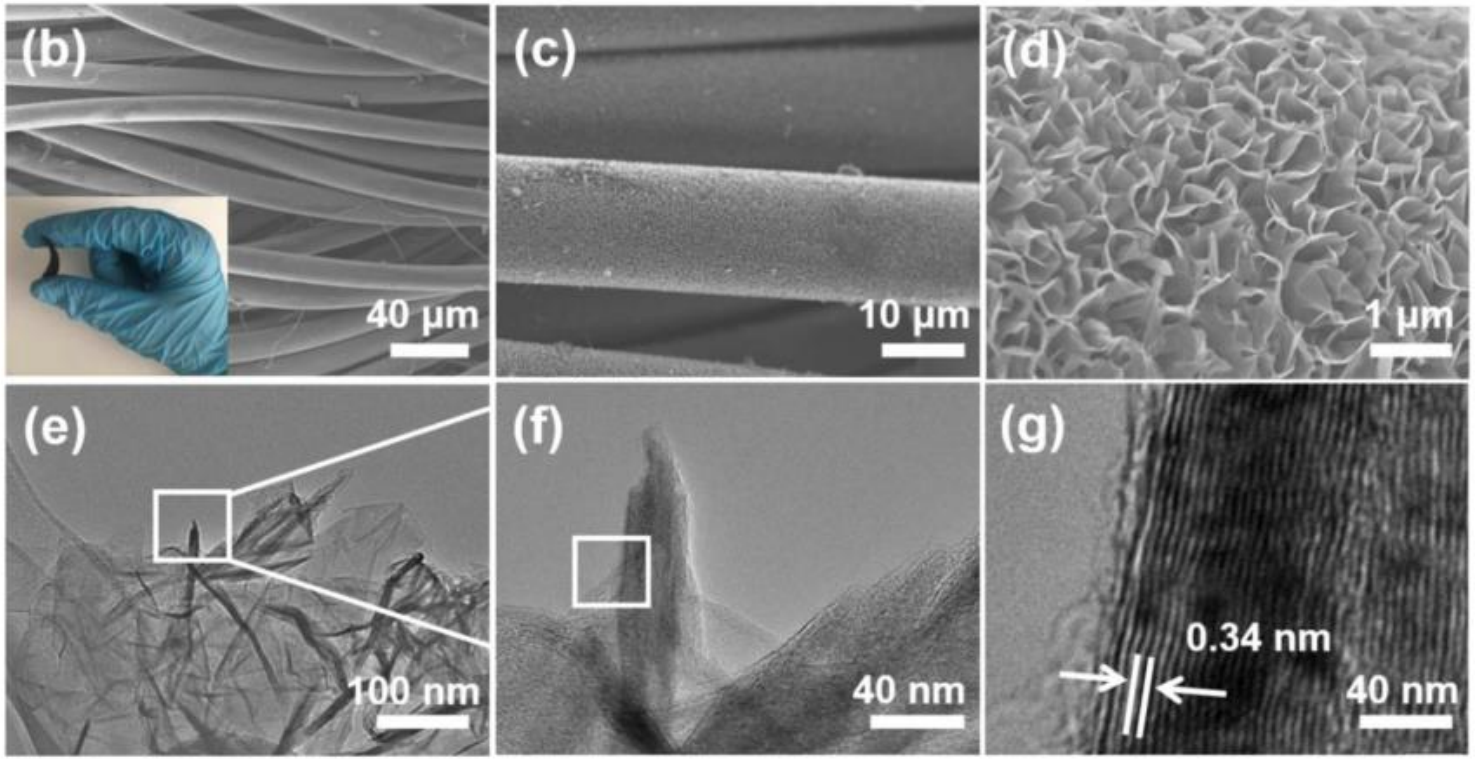
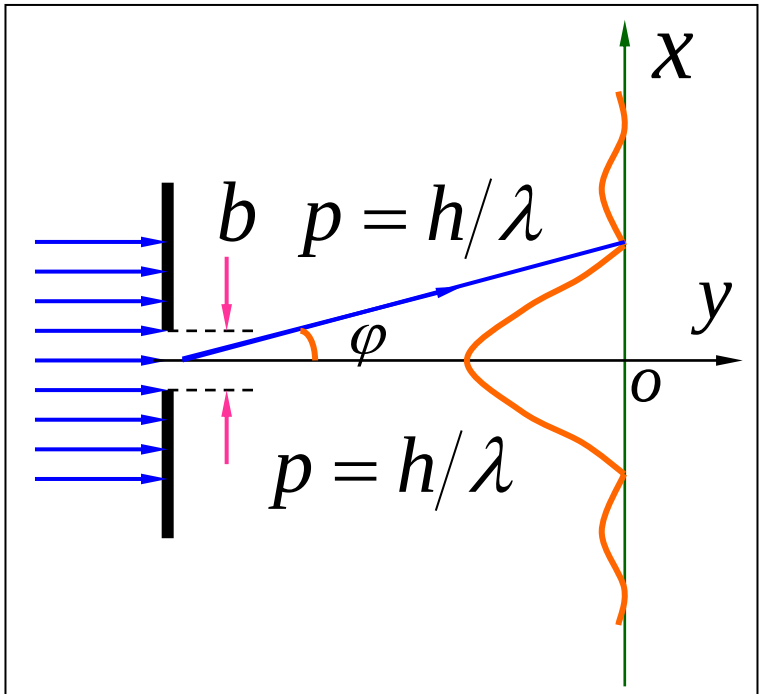


FIGURE 2 | (a) Schematic illustration of the synthesis process of the VG/CC by PECVD. **(b)** Low, **(c)** high, and **(d)** higher magnification SEM images of VG/CC. Insets: photograph of bent VG/CC, demonstrating the good mechanical strength of VG/CC. TEM image of VG/CC at **(e)** low and **(f)** high magnifications, **(g)** HRTEM images of VG/CC.

Front. Chem. 7 (2019)714.

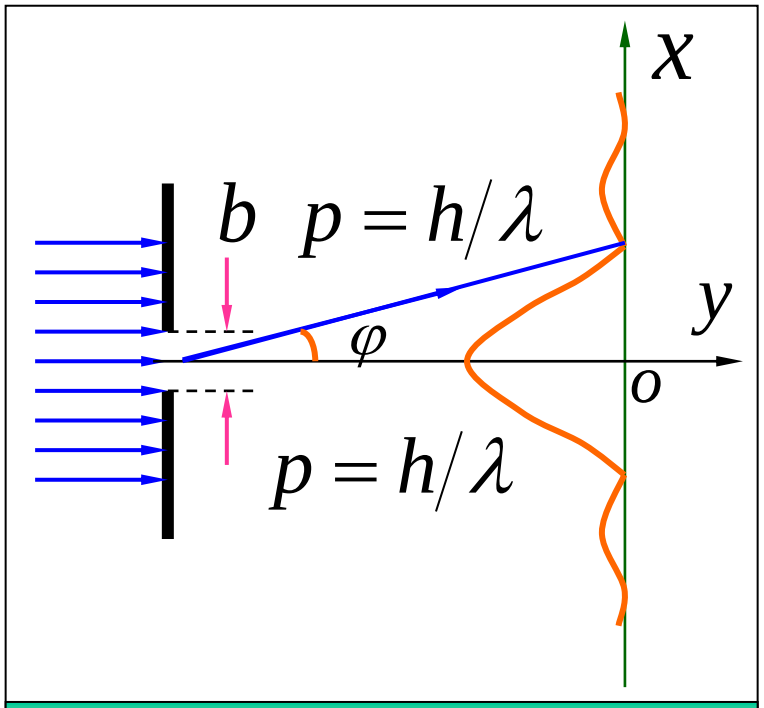
11.3.3 The Uncertainty Principle



Electron diffraction experiment

An incident beam of electrons is diffracted at the slit in screen, forming a typical diffraction pattern on screen. If the slit is made narrower, the pattern becomes wider.

11.3.3 The Uncertainty Principle



Electron diffraction experiment

The first minimum of the diffraction pattern occurs at an angle θ

$$\sin\theta = \frac{\lambda}{\Delta y}$$

If the angle θ is small enough, θ replace sinθ

$$\theta = \frac{h}{p\Delta y} \qquad \lambda = \frac{h}{p}$$

To reach the first minimum, θ must be:

$$\theta = \frac{\Delta p}{p}$$

Therefore:

$$\Delta y \bullet \Delta p_y \geq h$$

11.3.3 The Uncertainty Principle

W.K.Heisenberg，1901—1976



It is impossible to make these two measurements simultaneously with unlimited accuracy.

we can't describe the motions of electrons in a classical way finds expression in the uncertainty principle, enunciated by Werner Heisenberg (1901—1976) in 1927.

11.3.3 The Uncertainty Principle

The uncertainty principle states that if a measurement of position is made with uncertainty Δy and a simultaneous measurement of momentum is made with uncertainty Δp， then the product of the two uncertainties can never be less than a number on the order of h

$$\Delta y \bullet \Delta p_y \geq h$$

Uncertainty principle is a natural outgrowth of the wave packet model.

11.3.3 Example

Example 1. An electron has a speed of 300m s^{-1}, accurate to 0.010%. With what fundamental accuracy can we locate the position of this electron?

Solution:

The electron momentum is

$$p = mv = (9.1 * 10^{-31}\,\text{kg})(300\ \text{m s}^{-1}) = 2.7 * 10^{-28}\,\text{kg m s}^{-1}$$

The uncertainty in momentum is given to be 0.01%:

$$\Delta p = 0.0001 * 2.7 * 10^{-28}\,\text{kg m s}^{-1} = 2.7 * 10^{-32}\,\text{kg m s}^{-1}$$

The minimum uncertainty in position is:

$$\Delta y = \frac{h}{\Delta p} = \frac{h}{\Delta(mv)} = \frac{6.626 * 10^{-34}\,\text{Js}}{2.7 * 10^{-32}\,\text{kg m s}^{-1}} = 2.4cm$$

11.3.3 Example

Example 2. A bullet has a speed of 300m s^{-1}, accurate to 0.010%. With what fundamental accuracy can we locate its position? Its mass is 50 g (= 0.050 kg)

Solution:

The electron momentum is

$$p = mv = (0.05\text{kg})(300\ \text{m s}^{-1}) = 15\text{kg m s}^{-1}$$

The uncertainty in momentum is given to be 0.01%:

$$\Delta p = 0.0001 * 15\text{kg m s}^{-1} = 1.5 * 10^{-3}\ \text{kg m s}^{-1}$$

The minimum uncertainty in position is:

$$\Delta y = \frac{h}{\Delta p} = \frac{h}{\Delta(mv)} = \frac{6.626 * 10^{-34}\ \text{Js}}{1.5 * 10^{-3}\ \text{kg m s}^{-1}} = 4.4 * 10^{-31}\ \text{m}$$

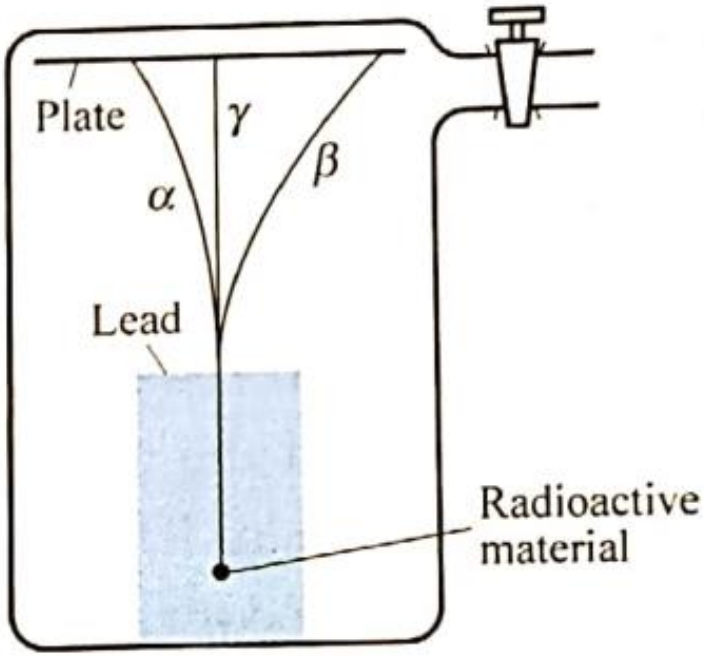
11.4 Radioactivity



Marie Curie was a physicist and chemist of Polish upbringing and, subsequently, French citizenship. She was a pioneer in the field of radioactivity, the first person honored with two Nobel Prizes, and the first female professor at the University of Paris.

Her achievements include the creation of a theory of radioactivity (a term coined by her), techniques for isolating radioactive isotopes, and the discovery of two new elements, polonium and radium. It was also under her personal direction that the world's first studies were conducted into the treatment of neoplasms ("cancers"), using radioactive isotopes.

11.4 Radioactivity



Unstable nuclei spontaneously decay by breaking apart or rearranging their internal structures in a process called radioactivity.

The three emanations from a radioactive material and their paths in a magnetic field perpendicular to the plane of the diagram. Naturally accruing radioactivity produces

(1) α (alpha particles): positive charged
(2) β (beta particles): negative charged
(3) γ (gamma rays): neutral

11.4 Radioactivity

Alpha (a) rays consists of positively charged particles, each particle being the ${}_{2}^{4}He$ nucleus of helium.

The most common kind of beta (β) rays consists of negatively charged particles, or β^- particles, which are electrons. Another kind of β ray consists of positively charged particles, or β^+ particles. A β^+ particle, also called a positron, has the same mass as an electron but carries a charge of +e instead of -e.

Gamma (γ) rays are high-energy photons. If a radioactive parent nucleus disintegrates into a daughter nucleus that has a different atomic number, one element has been converted into another element, the conversion being referred to as a transmutation of elements.

11.4.2 Radioactive decay law

The number of radioactive nuclei in any sample of radioactive material decreases continuously

$$\frac{dN}{dt} = -\lambda N$$

$$N = N_0 \mathbf{e}^{-\lambda t}$$

N_0: the number of nuclei at t=0

11.4.2 Radioactive decay law

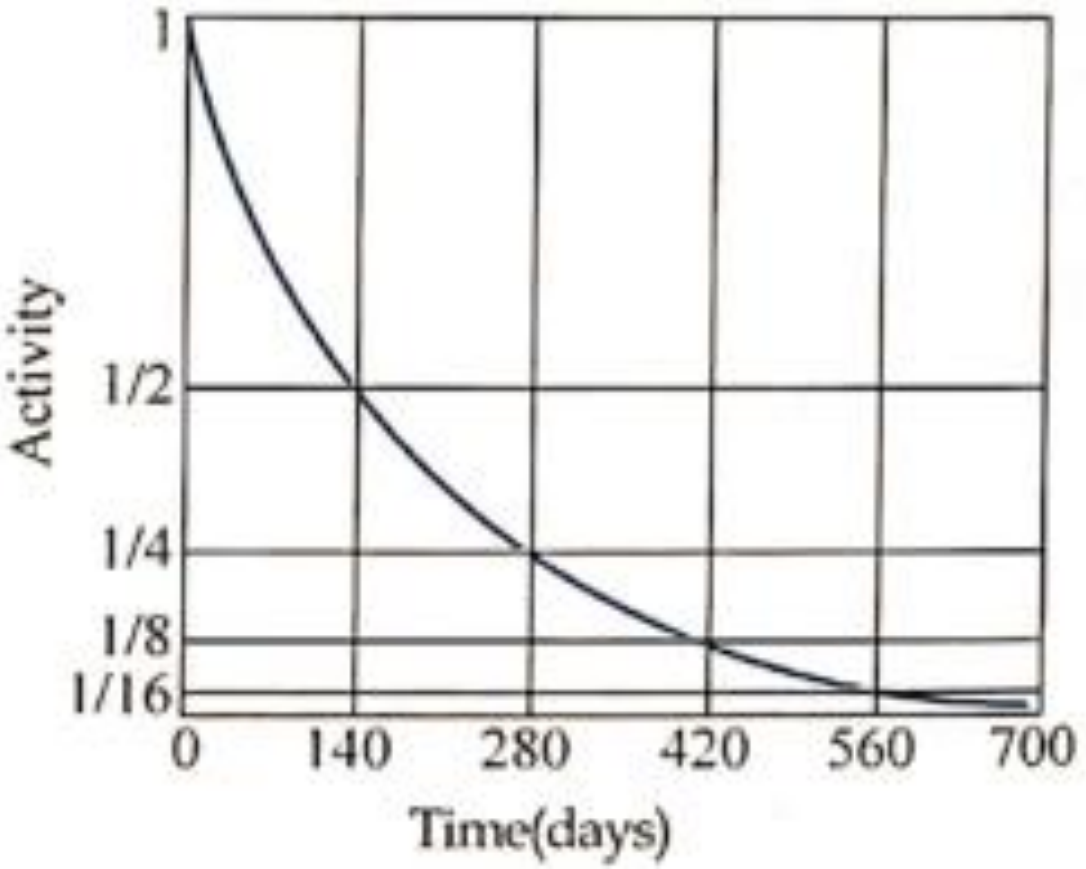
The half-life $T_{1/2}$ of a radioactive sample is defined as the time at which the number of radioactive nuclei has decreased to one-half the number at t = 0.

$$e^{-\lambda T_{1/2}} = 1/2$$

$$\lambda T_{1/2} = \ln 2$$

$$T_{1/2} = \ln 2/\lambda = 0.693/\lambda$$

11.4.2 Radioactive decay law



Decay curve for the radioactive element polonium. Polonium has a half-life of 140 days.

11.4.2 Radioactive decay law

When $t=1/\lambda$

$$N=N_0 e^{-\lambda t}$$

$$N=N_0 e^{-1}=\frac{1}{e} N_0$$

Therefore, λ is the time to decrease to 1/e of its initial value

The half-life of selected radioactive atoms has been used to determine the age of the artifacts and the age of geologic sample.

$$^{14}_{6}C \rightarrow {}^{14}_{7}N + e + \nu$$

The half-life time for this decay is 5730 years.

11.5 Elementary particles

Table 11-2 Some Particles and Their Properties

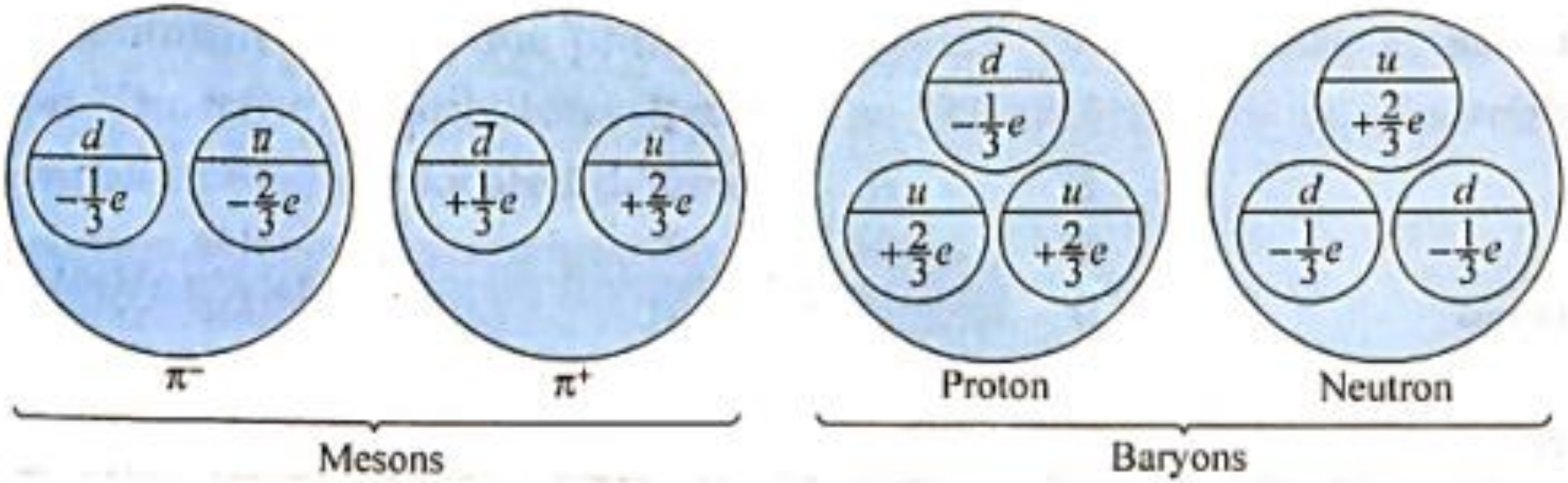
Family	Particles	Particle Symbol	Antiparticle Symbol	Rest Energy/MeV	Lifetime/s
Photon	Photon	γ	Self	0	Stable
Lepton	Electron	e^- (or β^-)	e^+ (or β^+)	0.511	Stable
	Muon	μ^-	μ^+	105.7	2.2×10^{-6}
	Tau	τ^-	τ^+	1784	10^{-13}
	Electron neutrino	ν_e	$\bar{\nu}_e$	≈0	Stable
	Muon neutrino	ν_μ	$\bar{\nu}_\mu$	≈0	Stable
	Tau neutrino	ν_τ	$\bar{\nu}_\tau$	≈0	Stable
Hardron					
Mesons					
	Pion	π^+	π^-	139.6	2.6×10^{-8}
		π^0	Self	135.0	0.8×10^{-16}
	Kaon	K^+	K^-	493.7	1.2×10^{-8}
		K_S^0	$\bar{K}_S^0$	497.7	0.9×10^{-10}
		K_L^0	$\bar{K}_L^0$	497.7	5.2×10^{-8}
	Eta	η^0	Self	548.8	$<10^{-18}$
	⋮				
	Plus other mesons				
Baryons					
	Proton	$\bar{p}$	$\bar{p}$	938.3	Stable
	Neutron	n	$\bar{n}$	939.6	900
	Lambda	Λ^0	$\bar{\Lambda}^0$	1116	2.6×10^{-10}
	Sigma	Σ^+	$\bar{\Sigma}^-$	1189	0.8×10^{-10}
		Σ^0	$\bar{\Sigma}^0$	1192	0.6×10^{-20}
		Σ^-	$\bar{\Sigma}^+$	1197	1.5×10^{-10}
	Omega	Ω^-	Ω^+	1672	0.8×10^{-10}
	⋮				
	Plus other baryons				

11.5 Elementary particles

Table 11-3 Quarks and Antiquarks

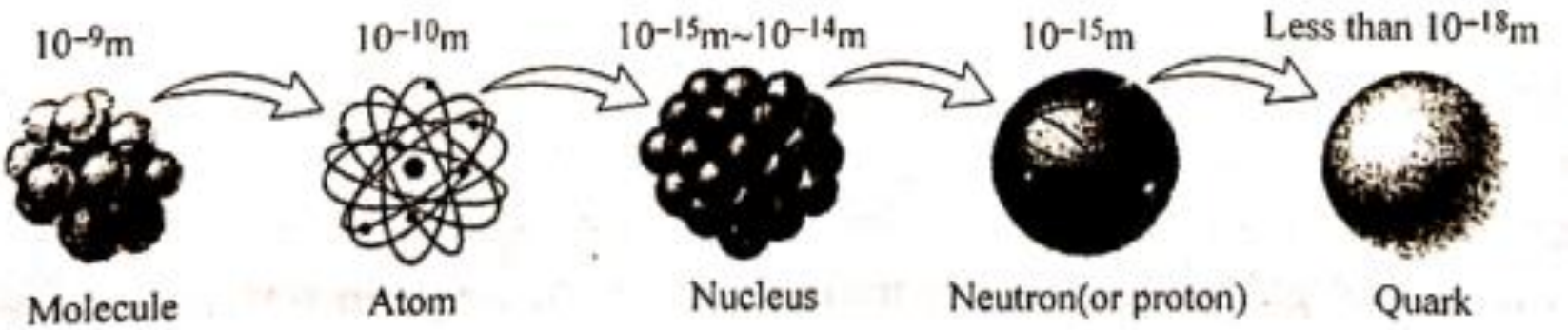
Name	Quarks		Antiquarks	
	symbol	Charge	Symbol	Charge
Up	u	$+\frac{2}{3}e$	$\bar{u}$	$-\frac{2}{3}e$
Down	d	$-\frac{1}{3}e$	$\bar{d}$	$+\frac{1}{3}e$
Strange	s	$-\frac{1}{3}e$	$\bar{s}$	$+\frac{1}{3}e$
Charmed	c	$+\frac{2}{3}e$	$\bar{c}$	$-\frac{2}{3}e$
Top	t	$+\frac{2}{3}e$	$\bar{t}$	$-\frac{2}{3}e$
Bottom	b	$-\frac{1}{3}e$	$\bar{b}$	$+\frac{1}{3}e$

11.5 Elementary particles



According to the original quark model of hadrons, all mesons consist of a quark and an antiquark, while baryons contain three quarks.

11.5 Elementary particles



Matter is composed of basic units, starting with a molecule and ending with a quark.

Homework

Page 279， **9， 14, 15, 18, 20.**

Homework file name:

Roll number+week 15 + name,

for example: 2023800901XX + week 15 + Alfa Robot

pdf file, send to the cloud)

Questions

1. In analyzing the photoelectric effect, how can we be sure that each electron absorbs only one photon?

2. Can you devise an experiment to measure the work function of a material?

3. Would the photoelectric effect be useful in a burglar alarm system? In a smoke system?

Questions

4. Is charge quantized in physics? Is mass quantized? Describe some experiments that support your answer.

5. Photon A has twice the energy of photon *B*. What is the ratio of the momentum of A to that of *B?*

6. In the photoelectric effect, why does the existence of a cutoff frequency (threshold frequency) speak in favor of the photon theory and against the wave theory?

7. In Compton scattering, why would you expect to be independent of the material of which the scatter is composed?

Questions

8. Why don’t we observe a Compton effect with visible light?

9. What determines the shortest and longest wavelengths that a hydrogen atom can emit?

10. Is the Bohr model of the hydrogen atom consistent with the uncertainty principle?

Questions

11. Is the wave nature of electrons significant in the function of a television picture tube? For example, do diffraction effects limit the sharpness of the picture?

12. A proton and an electron have the same speed. Which has longer wavelength?

13. A proton and an electron have the same kinetic energy. Which has shorter wavelength?

14. Why is the wave nature of matter not more apparent to our daily observations?

Questions

15. How many experiments can you recall that support the wave theory of light? The particle theory of light? The wave theory of matter? The particle theory of matter?

16. Discuss similarities and differences between a matter wave and an electromagnetic wave?

17. In your body, are there more neutrons than protons? More protons than electrons? Discuss.

18. Why do particle physicists want to accelerate particles to higher and higher energies?

Homework

Please submit your homework , and Monitors send to me before 11-Jan-2024, 10 pm of Beijing Time.

Page 279， Q1, 3, 6,7， 8

1. A photoelectric surface has a work function of 4.0eV. What is the maximum velocity of the photoelectrons emitted by light of frequency 3×10^{15} Hz?

3. The photoelectric work function of potassium is 2.0eV. If light having a wavelength of 360nm falls on potassium, find

(a) the stopping potential difference,

(b) the kinetic energy in electron volts of the most energetic electrons ejected,and

(c) the velocities of these electrons.

6. (a) Show that the energy E of a photon (in eV) is related to its wavelength λ (innm) by

$$E = \frac{1240}{\lambda}$$

(b) hat is the energy in electron volts of a photon having a wavelength of 91.2nm?

7. A particular X-ray photon has a wavelength of 0.035nm. Calculate the photon's (a) energy, (b) frequency, and (c) momentum.

8. A special kind of light bulb emits monochromatic light at a wavelength of 600nm. It is rated at 60W, and is 90% efficient in converting electric energy to light. How many photons will the bulb emit over its 730h lifetime?

9，14, 15, 18, 20

9. Calculate the fractional change in the wavelength of an X-ray of wavelength 0.40Å (0.040nm) that undergoes a 90° Compton scattering from an electron.

14. (a) What is the least amount of energy in electron volts that must be given to a hydrogen atom so that it can emit the H_β line in the Balmer series?

(b) How many different possibilities of spectral line emission are there for this atom when the electron goes from $n_i = 4$ to the ground state?

15. An atom absorbs a photon having a wavelength of 370nm and immediately emits another photon having a wavelength of 580nm. How much net energy was absorbed by the atom in this process? (Ease the computation by using the result of problem 6.)

18. Using the classical relation between momentum and kinetic energy, show that the de Broglie wavelength of an electron can be written (a) as

$$\lambda = \frac{1.228\text{nm}}{\sqrt{K}}$$

in which K is the kinetic energy in electron volts, or (b) as

$$\lambda = \sqrt{\frac{1.51}{V}}$$

where λ is innm, and V is the accelerating potential difference in volts.

20. The uncertainty in momentum of an electron with a kinetic energy of approximately 25eV is 10 percent. What is the minimum uncertainty in its position?